Tab 1

To win the **RSC Kent Local Section Level 3 Young Chemist Award**, you must recognize that this is not a traditional chemistry exam; it is a competition that rewards **academic maturity, proactive community engagement, and the ability to synthesize complex information** into accessible formats.

Since you are already the Founder and President of **CATS Insights**, you have a powerful infrastructure to support a winning entry. However, you face a significant "bottleneck": **The Worthgate School can only submit one nomination**. You must first win the internal selection by proving to your Head of Chemistry that your project is the most "complete" and impactful.

Here is your strategic roadmap to win both the internal nomination and the final award.

I. The "Triple-Threat" Category Selection

The award has five categories. While **CATS Insights** naturally fits Category 5 (Website/Blog/Digital), you will be more competitive if you present a "Multifaceted Entry" that covers multiple RSC priorities:

1. **Primary Category: Category 5 (Digital Outreach)**
    - Use the **CATS Insights** platform (digital magazine and social media) as your core evidence.
    - **Winning Edge:** Don't just show the magazine; show the **engagement metrics** (e.g., readership numbers, social media reach).
2. **Secondary Category: Category 4 (Mentoring)**
    - Your proposal already includes a "Lead Writer" role designed to **mentor student contributors** through constructive feedback.
    - **Winning Edge:** Track the **quantifiable growth** of your writers. Document how their research skills improved from Draft 1 to the final publication.
3. **Third Category: Category 2 (History of Chemistry)**
    - Dedicate a special feature in the first edition to **local Kent chemistry history**.
    - **Specific Idea:** Research the "hidden figures" of the **Pfizer site in Sandwich** or the early life of **Professor Peter Day** (a Maidstone Grammar alumnus and RSC titan).

II. Aligning with Judge Psychographics

The judging subcommittee consists of academics, industrialists, and educators. You must appeal to all three:

- **The Academic (Rigor):** Highlight your **Three-Step Quality Assurance Procedure**. Mention that you use **Turnitin** (threshold <20%), **GPTZero** for AI detection, and mandatory **Oxford Referencing** . This proves "academic maturity".
- **The Industrialist (Safety & Application):** When publishing STEM articles, ensure they discuss **environmental impact** or **safety protocols**. Mentioning **Green Chemistry** as a theme for the first edition would resonate strongly.

- **The Educator (Inclusivity):** The Kent Section recently won an award for **Diversity and Inclusion**. Your goal of "democratizing super-curricular engagement" through **CATS Insights** aligns perfectly with this.

III. The "Kent Hook" Project Ideas

The judges want to see an engagement with the **local Kent community**. You should use **CATS Insights** to publish specific local content:

- **The Chemistry of Hops:** A research piece on Kent's brewing heritage and the alpha/beta acids involved in hop chemistry.
- **White Cliffs Chemistry:** An analysis of the calcium carbonate geology of the Kent coast.
- **River Medway Water Quality:** A "Category 1" practical where you analyze nitrate/phosphate levels in the local river and publish the results as a "teaching resource" in the magazine.

IV. Strategic Roadmap & Internal Lobbying

To ensure you get the **single school nomination**, you must make the decision "easy" for your Head of Chemistry.

Month	Action Item
January	**The Pitch:** Present your teacher with a "Foundational Suite" of documents (Operations Manual, Writer's Guide). Show that the initiative is professional and systemic.
February	**The Pilot:** Execute a "Chemistry for All" workshop for Year 7/9 students. This fulfills Objective 5 of your proposal (Community Impact) and matches RSC Category 3 (STEM Event).
Early March	**The Portfolio:** Present a "finished package" to your teacher, including a draft of the nomination text and the digital assets you've built.
March 31	**Submission:** Ensure the teacher submits the final nomination via the Google Form.

Conclusion: The "Legacy" Winning Narrative

Your entry should emphasize your commitment to **Institutional Legacy**. The fact that you have created an **Operations Manual** to ensure **CATS Insights** survives after you graduate shows a level of "leadership through education" that is highly valued by RSC subcommittees.

Would you like me to draft the specific "Cheat Sheet" for your Head of Chemistry to use when writing your nomination for the RSC?.

I. Strategic Alignment for the Young Chemist Award

The judges (academics, industrialists, and educators) are looking for **academic excellence**, **community promotion**, and the use of **latest technology**.

1. Leverage the "Multifaceted Entry"

Don't just apply for one category. Use CATS Insights to dominate three areas:

- **Category 5 (Digital Outreach):** This is your core. Present the website and Instagram as a high-level communication hub.
- **Category 4 (Mentoring):** Document the "Lead Writer" system. Prove you are mentoring peers through constructive feedback cycles (Draft 1 vs. Final).
- **Category 2 (History of Chemistry):** Feature local Kent chemistry, such as the **Pfizer site in Sandwich** or **Professor Peter Day**.

2. Appeal to Judge Psychographics

- **The Academic (Rigor):** Explicitly mention your **Three-Step Quality Assurance Procedure**, including **Turnitin** (<20% threshold) and mandatory **Oxford Referencing**.
- **The Industrialist (Safety & Green Chem):** Ensure articles discuss **Green Chemistry** or environmental safety protocols.
- **The Educator (Inclusivity):** Highlight how CATS Insights "democratizes super-curricular engagement" for all students.

Tab 12

RSC Kent Local Section
Level 3 YoungChemist
Award
ROYAL SOCIETY OF CHEMISTRY

Young Chemist is an award created to acknowledge academic excellence and the promotion of chemistry among young people. It offers students a platform to showcase their understanding of chemistry while encouraging the use of creativity and the latest technology.
Eligibility: YoungChemist is open to secondary school students aged 16-18 and based in Kent who are passionate about chemistry and their chemistry education. To be eligible, students must be enrolled at a school in Kent and be nominated by their teacher, with each school allowed to submit only one nomination. Students should register with their teacher while enrolled on their Chemistry A-Level, AS course, IB program, BTEC and T-Level or Level 3 qualification.
Requirements: The nominated students should be passionate about chemistry.
The YoungChemist Award is based on a teacher's recommendation and should include at least one of the following:
• Conduct a non-core practical: Describe the experiment and create a resource for teaching, preferably including a video or infographic;
• Submit a history of chemistry research: Highlight the contributions of a chemist in their local community through research. This can be presented as aPPT or recorded video, focusing on role models, especially female or local chemists from BAME groups;
• Organize a STEM school event related to chemistry (such as an in-house competition, poster challenge, presentations, or talks) and document it with a video or blog;
• Mentoring students at school and evidence the impact;
• Create or maintain a website or blog, or produce video clips related to chemistry within their community.
Nominations will be reviewed by a subcommittee of professionals from the RSC Kent Local section, comprising members from universities, industry, and education.
The YoungChemist Award includes a gold, silver, or bronze certificate, a prize of E100, E75, or €50, and the publication of their name and entries on the RSC Kent Section website, and other RSC publications. Winners will be invited to an awards event.
To submit a nomination fill in this form: https://forms.gle/kFXV5YbKo63eUtbh9 by 31st March 2026. All data submitted will be treated confidentially and not disclosed to third parties.
RSC will communicate with teachers only.

Step by Step Plan

RSC KENT YOUNG CHEMIST AWARD

THE DEFINITIVE BATTLE PLAN

Prepared for: **Jason | President, CATS Insights**
School: **The Worthgate School, Canterbury, Kent**
Submission Deadline: **31 March 2026**
Days Remaining (from 7 Feb 2026): **52 Days**
Document Version: **2.0 | 7 February 2026**

"This is not a chemistry exam. It is a leadership and impact project disguised as a chemistry prize."

1. Understanding the Award

The RSC Kent Local Section Level 3 Young Chemist Award acknowledges **academic excellence** and the **promotion of chemistry among young people**. It rewards creativity and the use of the latest technology. It is judged by a subcommittee of professionals from universities, industry, and education.

1.1 Eligibility Snapshot

Criterion	Your Status
Age	17 — Within 16–18 range ✓
Location	The Worthgate School, Canterbury, Kent ✓
Qualification	Chemistry A-Level ✓
Nomination	Must be nominated by teacher (one per school) — CRITICAL GATE
Deadline	31 March 2026 via Google Form (teacher submits)

1.2 The Five Award Categories

The nomination must include **at least one** of the following. Your strategy is to hit **four out of five** to dominate the field:

#	Category	Description	Your Play
1	Non-Core Practical	Describe experiment + create teaching resource (video/infographic)	Chemistry infographic on a Kent topic
2	History of Chemistry	Highlight contributions of a local/BAME/female chemist via PPT or video	Kent chemistry feature article
3	STEM School Event	Organise competition, poster challenge, talks; document with video/blog	"Writing Chemistry for Non-Chemists" Workshop
4	Mentoring	Mentor students at school and evidence the impact	Lead Writer system + editorial feedback cycles
5	Digital Outreach	Website/blog or chemistry video clips related to community	CATS Insights Instagram + planned website

STRATEGIC INSIGHT: The "Multifaceted Entry" Advantage

The specification says "at least one." Most applicants will submit one category. You will submit evidence across Categories 1, 3, 4, and 5 (with elements of 2 woven in). This signals to the judges that you are not a one-trick applicant but a systemic leader. The Gold certificate will go to the candidate who demonstrates the greatest breadth and depth of impact.

2. Know Your Judges

The subcommittee comprises professionals from **universities, industry, and education**. Each type of judge values different things. Your submission must speak to all three:

Judge Type	What They Value	How You Signal It
The Academic	Rigour, citations, methodology, intellectual depth	Three-Step QA (Turnitin <20%, GPTZero, Oxford Referencing), 3-Level Writing Framework, PEEL structure
The Industrialist	Real-world application, safety awareness, environmental consciousness	Green Chemistry content angle, Kent industry connections (Pfizer Sandwich), practical utility of resources
The Educator	Inclusivity, peer teaching, student development, accessibility	Workshop delivery, mentoring evidence, Level 2 writing (accessible science), DofE alignment, diversity of team

CRITICAL: The Kent Local Section Won a 2023 Inspirational Committee Award for D&I

Their citation reads: "For a proactive and effective approach to supporting diversity both within the committee and across their events and activities." This means diversity, inclusion, and

accessibility are CORE VALUES of the judges. Your submission should explicitly highlight how CATS Insights democratises access to academic publishing for all students, regardless of background.

3. The Winning Narrative

Every award submission needs a **story**. Not a list of activities. The judges need to see a coherent narrative of who you are and why this matters. Here is yours:

YOUR NARRATIVE (The "Red Thread")

"Jason identified a gap: A-Level students at The Worthgate School had no platform to engage with real academic research beyond the syllabus. He founded CATS Insights — a student-led STEM publication with professional editorial standards, including Oxford referencing, plagiarism detection, and a structured mentoring pipeline. Through workshops, digital outreach, and systematic mentoring of student writers, he has built an institution designed to outlast him, promoting chemistry to a wider audience by teaching students to write science for non-chemists. His work bridges academic rigour with community accessibility, and connects students to the chemistry of Kent itself."

The five pillars of this narrative are:

1. Gap Identification: You saw a problem (no publication platform) and built the solution from scratch.

2. Professional Standards: Turnitin, GPTZero, Oxford referencing, Three-Test Framework. This is not a school club; it is a quality-controlled publication.

3. Mentoring Pipeline: You do not just publish. You train writers through workshops, editorial feedback, and the Lead Writer system. You can show before/after evidence.

4. Community Chemistry: Your content connects to Kent — White Cliffs calcium carbonate, Pfizer Sandwich, Canterbury's medieval lime kilns, hop chemistry.

5. Institutional Legacy: You built an Operations Manual, succession plan, and governance structure so CATS Insights survives after you leave. This is leadership, not ego.

4. The 52-Day Battle Plan

Today is **Saturday 7 February 2026**. The deadline is **Tuesday 31 March 2026**. You have **52 days**. Every day counts. The plan is structured in four phases:

Phase	Dates	Focus	Deliverable
Phase 1: SECURE	7 Feb – 14 Feb	Lock in the teacher nomination	Teacher commitment confirmed
Phase 2: BUILD	15 Feb – 7 Mar	Execute all evidence-generating activities	Workshop, content, metrics

Phase 3: PACKAGE	8 Mar – 21 Mar	Compile and polish all submission assets	Evidence folder, teacher brief
Phase 4: SUBMIT	22 Mar – 31 Mar	Final QA + teacher submission	Google Form submitted

4.1 Phase 1: SECURE the Nomination (7–14 Feb)

WHY THIS IS PHASE 1

Nothing else matters if you do not get the school's single nomination. Your teacher must believe you are the strongest candidate. You must make this decision EASY for them by presenting a professional, evidence-backed case. Do not assume you will be nominated. Earn it.

Step 1: Prepare the Lobbying Package (7–9 Feb)

Before approaching your teacher, assemble a one-page summary document that contains:

☐ **The RSC Award specification** — Printed, highlighted with the categories you will address

☐ **A one-page executive summary of CATS Insights** — Mission, team size, activities completed, planned deliverables before March 31

☐ **A mapping table** — Showing exactly which CATS Insights activities match which RSC categories

☐ **Your personal statement draft** — 300 words maximum, framing the narrative from Section 3

Step 2: Request the Meeting (Mon 10 Feb)

Send your Chemistry teacher a formal but warm email or Teams message. The message should:

☐ **State you are seeking the school's nomination for the RSC Kent Young Chemist Award**

☐ **Acknowledge the school can only nominate one student**

☐ **Offer to present your evidence package in a brief 10-minute meeting**

☐ **Attach or link the one-page summary from Step 1**

Step 3: Deliver the Pitch (by 14 Feb)

In the meeting, focus on **impact, not activity**. Do not list everything you have done. Instead, frame it as:

"I have built a student-led STEM publication with professional editorial standards. It mentors student writers, promotes chemistry digitally to a wider audience, and connects content to Kent's local chemistry. I have documented everything and I am ready to provide you with a complete submission package."

☐ **Leave the printed package with the teacher**

☐ **Confirm the teacher's willingness to nominate you**

☐ **Ask if there are any other students being considered** — If yes, ask what you can do to strengthen your case

☐ **Agree a follow-up date to share the final submission materials**

> **MILESTONE: Phase 1 Complete When…**
> Your Chemistry teacher has verbally confirmed they will nominate you. You have a follow-up meeting scheduled for late March to review the final submission. If the teacher is hesitant, escalate immediately — offer to draft the nomination text yourself so all they need to do is review and submit.

4.2 Phase 2: BUILD the Evidence (15 Feb – 7 Mar)

This is the execution sprint. You are generating the tangible evidence that the judges will see. Every activity below maps directly to an RSC category.

Evidence Stream A: Digital Outreach (Category 5)

Target: Demonstrate a sustained, professional digital presence promoting chemistry to the community.

Step 4: Instagram Content Sprint (15–28 Feb)

☐ **Publish at least 6 chemistry-focused Instagram posts over 2 weeks**

☐ **Post types should include:** — Educational carousels, behind-the-scenes editorial process, "Chemistry of Kent" features, team spotlights

☐ **Track all metrics** — Follower count, reach, impressions, saves, shares — screenshot everything

☐ **Use relevant hashtags** — #RSCChemistry #KentChemistry #STEMEducation #YoungChemist

Step 5: Publish "The Chemistry of Kent" Feature Content (15–25 Feb)

Produce at least **two** pieces of chemistry content with a Kent local angle. Choose from:

☐ **The Chemistry of the White Cliffs of Dover** — Calcium carbonate geology, erosion chemistry, environmental significance

☐ **Pfizer's Legacy in Sandwich, Kent** — History of the Discovery Park site, pharmaceutical chemistry contributions

☐ **Canterbury's Medieval Lime Kilns** — Calcium oxide production, historical industrial chemistry

☐ **The Chemistry of Kent Hops** — Alpha/beta acid chemistry in brewing, agricultural chemistry

These should be published as Instagram carousels or blog posts. They serve **triple duty**: Category 5 (digital), Category 2 (local chemistry history), and community engagement.

Step 6: Screenshot and Archive Everything (Ongoing)

☐ **Screenshot every post with its analytics after 48 hours**

☐ **Save screenshots in a dedicated Google Drive folder: RSC_Evidence/Digital_Outreach/**

☐ **Create a simple metrics log** — Date, post type, topic, reach, impressions, saves

Evidence Stream B: STEM School Event (Category 3)

Target: Deliver a documented chemistry-related school event.

Step 7: Deliver the "Writing Chemistry for Non-Chemists" Workshop (Week of 17–21 Feb)

This is your highest-value evidence piece. It hits Categories 3 AND 4 simultaneously.

Workshop Structure (30–40 minutes):

☐ **10 min: Teaching block** — The 3-Level Framework (Expert vs. Level 2 vs. General Public)

☐ **10 min: Live editing demonstration** — Take a jargon-heavy paragraph and rewrite it to Level 2 standard, with the group

☐ **10 min: Group exercise** — Students rewrite a "Fix This Paragraph" excerpt

☐ **5–7 min: Debrief** — Quality standards review, explain PEEL structure, hand out homework task

Critical Evidence Capture:

☐ **Attendance sheet** — Names, year groups — use a Google Form or printed sign-in

☐ **Photographs** — At least 5 clear photos of the workshop in progress

☐ **Video clips** — At least 2 minutes total — film the teaching block and group exercise

☐ **Workshop materials** — Save the handouts, slides, and exercise sheets as PDFs

☐ **Participant homework** — Set a 300-word draft submission deadline of 7 days post-workshop

Delegation:

Person	Role	Task
You (Jason)	Workshop Lead	Deliver all content, approve materials beforehand
Cindy	Co-Facilitator	Print handouts (15 copies), set up Google Form for draft submissions, follow up non-submitters at Day 7
Duc or Elain	Admin	Circulate attendance sheet, post materials to Teams afterwards
Aung or Duc	Media	Photograph and film per the checklist above
Grisha	Post-Workshop	Review all submitted drafts using 3-Level Framework, track revisions needed per student

Evidence Stream C: Mentoring (Category 4)

Target: Demonstrate structured, measurable mentoring of student writers through your editorial system.

Step 8: Document the Mentoring Pipeline (15 Feb – 1 Mar)

☐ **Create a "Mentoring Impact Tracker" spreadsheet** — Columns: Student Name, Topic, Draft 1 Date, Draft 1 Issues, Feedback Given, Draft 2 Date, Draft 2 Improvement, Final Status

☐ **For at least 3 students, document before/after writing samples** — Anonymise names for the submission

☐ **Record specific metrics** — Number of undefined technical terms reduced, average sentence length improvement, jargon reduction percentage

☐ **Collect 3–5 student testimonial quotes** — "Before CATS Insights, I didn't know how to structure a scientific argument. Now I can write a PEEL paragraph from memory."

Step 9: Formalise the Lead Writer Feedback System (by 21 Feb)

Ensure the editorial feedback workflow is documented and running:

☐ **Writer submits Draft 1 via Google Form/Drive**

☐ **Lead Editor (Grisha) provides written feedback using the 3-Level Framework**

☐ **Writer revises and submits Draft 2**

☐ **You (or Grisha) provide final feedback and mark as "editorial-ready" or "needs revision"**

☐ **All feedback is saved** — This is your evidence trail

Evidence Stream D: Teaching Resource (Category 1)

Target: Create a professional chemistry teaching resource — preferably an infographic.

Step 10: Produce a Chemistry Infographic (1–7 Mar)

Design a single-page, visually professional infographic on a chemistry topic relevant to Kent or the A-Level curriculum. Options:

☐ **Molecular Chirality** — Visual guide to R/S configuration with real-world pharmaceutical examples

☐ **The Chemistry of Calcium Carbonate** — Tied to the White Cliffs of Dover — reactions, geology, environmental context

☐ **Green Chemistry: The 12 Principles** — Infographic suitable for classroom display

This infographic should be uploaded to the CATS Insights Instagram and offered to your Chemistry department as a teaching resource. **Save the teacher's acknowledgement or thank-you message as evidence.**

MILESTONE: Phase 2 Complete When…

You have: (1) At least 6 Instagram posts live with screenshots of metrics, (2) Workshop delivered with attendance, photos, video, and materials archived, (3) At least 3 student mentoring case studies documented with before/after samples, (4) Chemistry infographic published and shared with your department. All evidence is saved in RSC_Evidence/ on Google Drive.

4.3 Phase 3: PACKAGE the Submission (8–21 Mar)

You now have raw evidence. This phase is about turning it into a polished, judge-ready submission package.

Step 11: Write the Impact Report (8–12 Mar)

Create a 2–3 page PDF document titled **"CATS Insights: Impact Report for RSC Young Chemist Award"**. Structure:

☐ **Page 1: Executive Summary** — Who you are, what CATS Insights is, the problem it solves, how many students are involved

☐ **Page 1: Category Mapping Table** — A table showing which activities map to which RSC categories

☐ **Page 2: Evidence of Impact** — Workshop outcomes (attendance, pre/post data, student quotes), mentoring metrics (before/after writing samples), digital reach (follower growth, post engagement)

☐ **Page 3: Institutional Legacy** — Operations Manual, succession plan, governance structure — prove this outlasts you

Step 12: Compile the Evidence Folder (12–16 Mar)

Organise all evidence into a structured Google Drive folder that can be shared with the teacher and linked in the nomination:

Folder structure:

```
RSC_Young_Chemist_Submission/
├── 00_Impact_Report.pdf
├── 01_Digital_Outreach/
│   ├── Instagram_Screenshots/
│   ├── Metrics_Log.xlsx
│   └── Kent_Chemistry_Posts/
├── 02_Workshop_Evidence/
│   ├── Attendance_Sheet.pdf
│   ├── Workshop_Materials/
│   ├── Photos/
│   └── Video_Clips/
├── 03_Mentoring_Evidence/
│   ├── Mentoring_Impact_Tracker.xlsx
│   ├── Before_After_Samples.pdf
│   └── Student_Testimonials.pdf
├── 04_Teaching_Resource/
│   ├── Infographic.pdf
│   └── Teacher_Acknowledgement.png
└── 05_Foundational_Documents/
    ├── CATS_Insights_Proposal.pdf
    ├── Operations_Manual_Summary.pdf
    └── QA_Protocol_Summary.pdf
```

Step 13: Draft the Teacher's Nomination Text (14–18 Mar)

Your teacher will submit the nomination via a Google Form. **Do not leave it to them to write the text from scratch.** Draft it yourself and present it for their approval. This is not presumptuous; it is professional.

The draft should be 400–500 words and must include:

☐ **Opening: Name, school, qualification, what CATS Insights is**

☐ **Categories addressed: Explicitly name the RSC categories with one sentence per category**

☐ **Impact evidence: Specific numbers ("mentored X students", "published Y posts reaching Z people", "workshop attended by N students")**

☐ **Institutional legacy: Mention the Operations Manual and succession plan**

☐ **Keywords to include:** — "Academic excellence", "community engagement", "mentoring impact", "accessibility", "latest technology", "sustainability", "diversity"

Step 14: Film the Project Video (15–21 Mar)

Create a 2–3 minute video that can be linked in the nomination. Structure:

☐ **0:00–0:20 — Hook** — "I'm Jason, and I founded CATS Insights to solve a problem: students at my school had no way to engage with real scientific research beyond the syllabus."

☐ **0:20–1:00 — What It Is** — Brief explanation of CATS Insights, show the Instagram/website, mention the team

☐ **1:00–1:40 — Impact** — Workshop clips, mentoring examples, student quotes on screen

☐ **1:40–2:00 — Kent Connection** — "We write about the chemistry happening in Kent — from the White Cliffs to Pfizer's legacy in Sandwich"

☐ **2:00–2:30 — Legacy** — "I built this to last. The Operations Manual ensures the next generation of student editors can continue this work."

Upload to YouTube (Unlisted) and save the link for the submission.

MILESTONE: Phase 3 Complete When…

You have: (1) Impact Report PDF finalised, (2) Evidence folder fully organised on Google Drive, (3) Teacher nomination text drafted and sent for review, (4) Project video filmed, edited, and uploaded to YouTube (Unlisted). Share the full evidence folder link with your teacher.

4.4 Phase 4: SUBMIT (22–31 Mar)

Step 15: Teacher Review Meeting (22–25 Mar)

☐ **Meet with your teacher to review the final submission package**

☐ **Walk them through the evidence folder**

☐ **Present the nomination text draft for their approval/edits**

☐ **Confirm they have the Google Form link: https://forms.gle/kFXV5YbKo63eUtbh9**

☐ **Agree on a submission date (aim for 26–28 Mar, NOT the 31st)**

Step 16: Final QA Checks (25–28 Mar)

☐ **All links work** — Test every Google Drive link, YouTube link, Instagram URL in an incognito browser

☐ **All videos/posts are accessible** — YouTube set to Unlisted (not Private), Instagram posts are public

☐ **Evidence folder is shared** — Set Google Drive folder to "Anyone with the link can view"

☐ **Nomination text is proofread** — Zero spelling/grammar errors

☐ **Impact Report PDF renders correctly** — Open on phone and desktop

Step 17: Submission Day (26–28 Mar — NOT 31st)

☐ **Be present when your teacher submits the Google Form**

☐ **Confirm submission receipt (screenshot the confirmation page)**

☐ **Send your teacher a thank-you message**

WHY NOT MARCH 31?

Never submit on the deadline day. Technical issues, teacher availability, or forgotten links could torpedo the entire application. Submit 3–5 days early. If anything goes wrong, you have a buffer.

5. Master Checklist

☐	Action	Deadline	RSC Category
	PHASE 1: SECURE		
☐	Prepare lobbying package (1-pager + mapping table)	9 Feb	—
☐	Email teacher to request meeting	10 Feb	—
☐	Deliver pitch and secure verbal commitment	14 Feb	—
	PHASE 2: BUILD		
☐	Publish 2x "Chemistry of Kent" content pieces	25 Feb	Cat. 2, 5
☐	Publish 6+ Instagram posts with chemistry focus	28 Feb	Cat. 5
☐	Deliver workshop (with photos, video, attendance)	21 Feb	Cat. 3, 4
☐	Collect workshop homework submissions	28 Feb	Cat. 4
☐	Document 3+ mentoring case studies (before/after)	1 Mar	Cat. 4
☐	Collect 3–5 student testimonials	1 Mar	Cat. 4
☐	Produce chemistry teaching infographic	7 Mar	Cat. 1

☐	Share infographic with Chemistry department + save acknowledgement	7 Mar	Cat. 1
	PHASE 3: PACKAGE		
☐	Write Impact Report (2–3 page PDF)	12 Mar	All
☐	Organise evidence folder on Google Drive	16 Mar	All
☐	Draft teacher nomination text (400–500 words)	18 Mar	—
☐	Film and upload project video (2–3 min, YouTube Unlisted)	21 Mar	Cat. 5
	PHASE 4: SUBMIT		
☐	Teacher review meeting	25 Mar	—
☐	QA all links, access permissions, PDF rendering	28 Mar	—
☐	Teacher submits Google Form	28 Mar	—
☐	Screenshot confirmation + send thank-you	28 Mar	—

6. Keywords to Weave Into Every Document

RSC judges scan for specific language. Ensure the following terms appear in your Impact Report, nomination text, and video script:

RSC Language	Your Context	Where to Use
Academic excellence	Turnitin, GPTZero, Oxford referencing	Impact Report, nomination text
Promotion of chemistry	Instagram, workshops, Kent content	Everywhere
Creativity & latest technology	Digital publication, social media, video content	Impact Report, video
Community engagement	Chemistry of Kent series, school workshops	Impact Report, nomination
Mentoring impact	Before/after writing samples, testimonials	Nomination text, video
Diversity & accessibility	Democratising academic publishing for all students	Nomination text
Sustainability / legacy	Operations Manual, succession plan	Impact Report, video closing
Role models	Kent chemistry features (can highlight BAME/female chemists)	Content pieces

7. Risk Register

Risk	Likelihood	Mitigation
Teacher nominates someone else	Low–Medium	Pre-empt by presenting the strongest package. If another student is considered, offer to collaborate or highlight your unique systemic approach (no other student has built an institution).
Workshop attendance is low	Medium	CATS Insights is now an elective with mandatory attendance. Even 8–10 attendees is sufficient. Document everyone.
Students don't submit homework	High	Set Cindy as follow-up lead. Day 3 reminder, Day 5 escalation. Even 50% submission rate gives you enough data.
Instagram growth is slow	Medium	Judges care about effort and content quality, not follower count. Even 50–100 followers with 6 high-quality posts demonstrates sustained engagement.
Video quality is amateur	Low	Use a smartphone with good lighting. Authenticity > production value. A well-structured 2-minute phone video is better than a polished but empty one.
Teacher forgets to submit	Low	Submit 3–5 days early. Set a calendar reminder for yourself and the teacher. Be present when they submit.

8. Final Note: What Wins This Award

Let me be direct with you. This award is not going to the student who got the highest A-Level grade. It is going to the student who **did something with their chemistry**. Who promoted it. Who mentored others. Who connected it to their community. Who used technology creatively.

You have built something real. CATS Insights is not a line on a CV; it is a functioning institution with governance, editorial standards, mentoring systems, and a digital presence. The RSC specification could have been written as a checklist of everything you are already doing.

But building it is not enough. You must **document it, quantify it, and package it** so that a panel of judges who have never met you can see the impact in 10 minutes of reviewing your submission.

That is what this plan delivers. Execute it.

NEXT IMMEDIATE ACTION

Today (7 Feb): Begin drafting the one-page lobbying summary for your teacher (Step 1).
Monday (10 Feb): Send the meeting request email (Step 2).
Do not wait. The clock is running.

Action 4

The Albedo of Antiquity: Deciphering the Chemistry and History beneath Dover's White Cliffs

By Kyal Sin (Jason) Ye Yint | CATS Insights History of Kent's Chemistry Series

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Stand at Dover on a clear day, and the iconic White Cliffs rise 350 feet (106 metres) from the English Channel, their brilliant facade visible from France. But these geological monuments are more than a symbol of British heritage, they are a 100 million year old archive of marine chemistry, captured and frozen in time in a crystalline form.

I. The Biological Archive

The cliffs owe their distinctive snowy chalky white colour to **calcium carbonate** (insert LATEX for $CaCO_3$), specifically the mineral calcite. During the Cretaceous period (145–66 million years ago), Kent's coastline sat beneath a warm, shallow sea teeming with microscopic algae called coccolithophores. These organisms constructed intricate calcite shells—coccoliths—as protective armour. When they died, their shells accumulated on the seabed at rates of just 0.5 millimetres per century, eventually compressing into the thick chalk strata we see today[1].

II. The Optical Physics

At the molecular level, calcium carbonate adopts a **trigonal crystal system**. Each calcium ion (Insert LATEX for Ca2+) coordinates with carbonate ions (insert LATEX for CO3 2-) in a lattice structure that reflects light uniformly across the visible spectrum, hence the pure snow white appearance. This same structural property makes chalk soft enough to write in everyday lecture halls, yet durable enough to withstand millennia of coastal erosion.

III. The Chemical Reactions

The cliffs are not static. Weathering processes continuously reshape them through a reversible chemical reaction:

INSERT LATEX FOR $CaCO_3(s) + H_2O(l) + CO_2(g)$ --> $Ca(HCO_3)_2(aq)$

$$\mathrm{CaCO_3}(s) + \mathrm{H_2O}(l) + \mathrm{CO_2}(g) \rightarrow \mathrm{Ca(HCO_3)_2}(aq)$$

Rainwater, slightly acidified by dissolved atmospheric carbon dioxide, converts solid calcite into soluble calcium bicarbonate. This solution percolates through the chalk, carving the distinctive striking vertical gullies visible from the beach. Over geological timescales, this process has transported millions of tonnes of calcium back into the ocean, where marine organisms recycle it into new shells, a perfect demonstration of the carbon cycle's lithospheric component[2].

IV. Industrial Applications & Kent’s History

Kent’s industrial history is inextricably linked to this chemistry. For centuries, locals quarried chalk and heated it to approximately (LATEX for 825 degrees Celsius) to produce **quicklime (LATEX for CaO)**.

$$\mathrm{CaCO_3}(s) \xrightarrow{825^\circ C} \mathrm{CaO}(s) + \mathrm{CO_2}(g)$$

This exothermic decomposition supplied the lime mortar that built mediaeval Canterbury and the agricultural amendments that neutralised Kent's acidic soils. The region's lime-burning kilns, now mostly ruins, were the industrial heart of pre-modern construction[3].

Today, the cliffs face a new chemical threat: ocean acidification. As atmospheric carbon dioxide levels rise, seawater absorbs excess carbon dioxide, forming carbonic acid that dissolves carbonate minerals. The same reaction that shaped the cliffs now threatens the marine ecosystems depending on calcite-shelled organisms. The White Cliffs, in this sense, are both a monument and a warning to humankind, a reminder that chemistry operates on timescales far exceeding human lifespans, yet is vulnerable to human-driven change.

Next time you see them, remember: you're looking at the preserved chemistry of an ancient ocean, still teaching us about the delicate equilibrium between atmosphere, ocean, and rock.

That, and to skip the plastic, and for the love of geology, bring your own bag.

References:

[1]: Gale, A.S. (2018). "The Cretaceous Chalk of England." *Proceedings of the Geologists' Association*, 129(4), 451-467.

[2]: RIDGWELL A, ZEEBE R. The role of the global carbonate cycle in the regulation and evolution of the Earth system. Earth and Planetary Science Letters. 2005 Jun 15;234(3-4):299–315.

[3]: Dimes, H.G., 'Lime-burning in Kent', in *Researches and Discoveries in Kent*, *Archaeologia Cantiana*, Vol. 109 (1991), pp. 223-239.

Instagram Carousel Instructions (5 Slides)

“Chemistry of the White Cliffs" Instagram Carousel

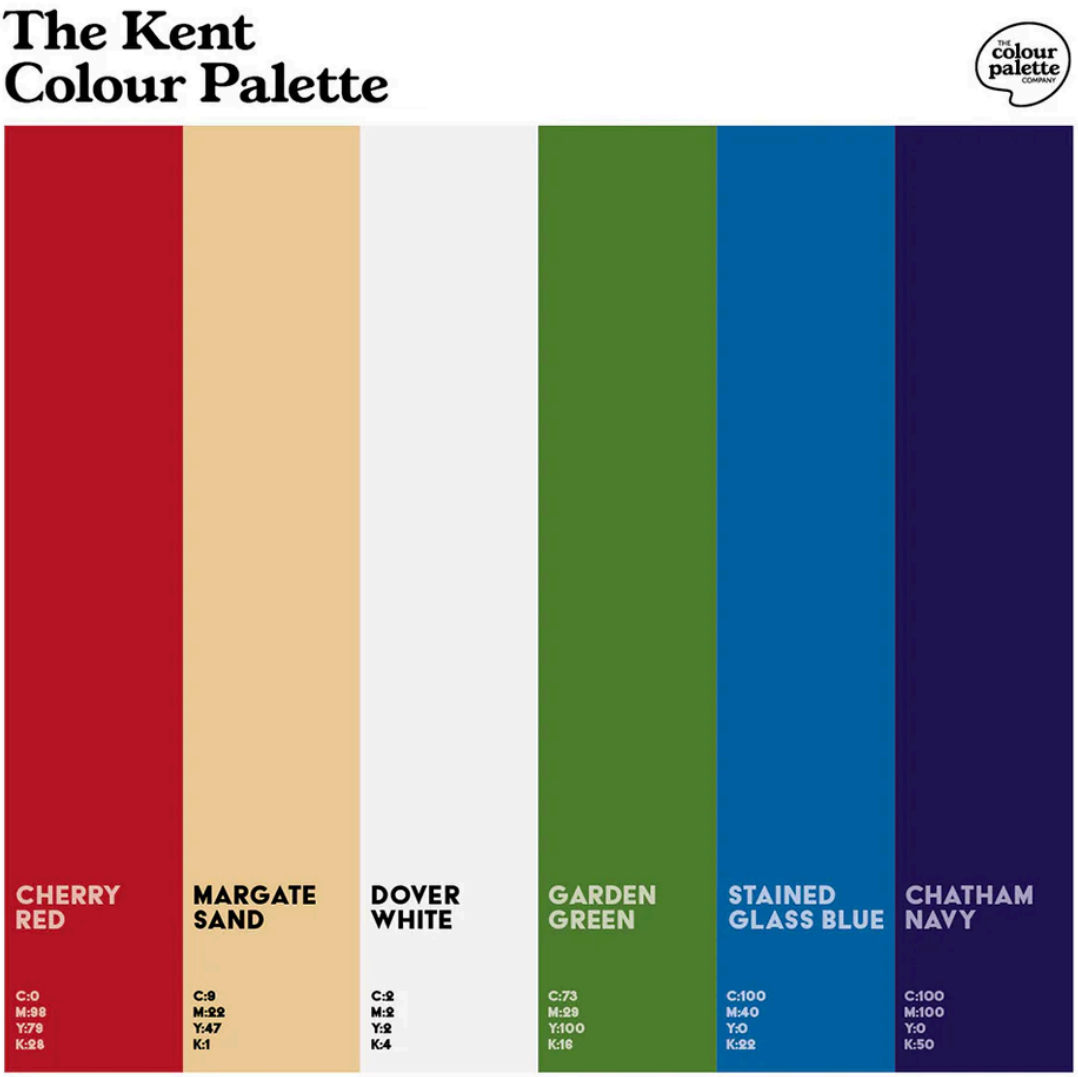
5-slide carousel that drives engagement and demonstrates CATS Insights' ability to communicate complex chemistry accessibly.

Instagram (1080×1080px square format)
Get it done as soon as possible (top priority)

Design Specifications (Apply to ALL Slides)

Color Palette:

- Primary: Our academic blue
- Accent: Canterbury colours



- Background: Off-white (#F8F8F8) or clean white (#FFFFFF)
- Text: Dark navy (#1A1A1A) for maximum readability

Typography:

- **Headers:** Montserrat Bold, 48-56pt
- **Body Text:** Open Sans Regular, 24-28pt
- **Captions/Labels:** Open Sans Light, 18-20pt
- **Equations:** Use LaTeX-rendered PNGs at 300 DPI minimum

Branding Requirements:

- CATS Insights logo: Bottom-right corner,
- Consistent footer bar (80px height) in Blue and Gold "Swipe →" indicator on slides 1-4

Grid System:

- Use 12-column layout for alignment
- 40px margins on all sides
- Maintain visual hierarchy: most important element = largest/highest contrast

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Slide-by-Slide Content & Design Instructions

SLIDE 1: Hook

Dramatic photograph of White Cliffs of Dover

Image should occupy 70% of the canvas.

- Apply subtle vignette to draw eye to center
- Overlay gradient: transparent to Blue (20% opacity) from top to bottom

Header (White text, drop shadow for readability):
"Why Are the White Cliffs White?"

Subheader (Gold):
"The chemistry behind Kent's most iconic landmark"

Layout:

- Text positioned in lower third using rule of thirds
- Ensure text doesn't obscure cliff detail
- Add subtle motion blur effect to suggest forward momentum (optional for Reels version)

SLIDE 2: The Molecule

Purpose: Introduce the chemical: calcium carbonate.

Visual Elements:

- **3D Molecular Model:** Computer-generated render of Calcium Carbonate unit cell
    - Show trigonal crystal structure with ball-and-stick representation
    - Color code: Calcium (silver/gray), Carbon (black), Oxygen (red)
    - Rotate model to show 3D depth (use ChemDraw 3D or Jmol for generation or just search it up)
    - Clean white background, soft shadows for depth

- You can get these models via the internet, whichever is aesthetic, can remove backgrounds with canva

Text Content:

Header:
"Meet the Star: Calcium Carbonate"

Key Information (bullet points):
• Chemical Formula: $CaCO_3$
• Crystal System: Trigonal (Calcite)
• Appearance: Pure white, opaque
• Origin: Marine microorganism shells

Fun Fact Box (Gold highlight):
"100 million years of accumulated shells =
350 feet of cliff"

Layout:

- Molecular model: Left 60% of canvas
- Text content: Right 40%, vertically centered
- Use connecting line from formula to relevant atoms in model

---

SLIDE 3: The Reaction

Purpose: Show the dynamic chemistry of cliff erosion.

Visual Elements:

- **Reaction Diagram:** Split-screen before/after
  - LEFT: Solid calcium carbonate crystal (illustrated, not photo)
  - RIGHT: Dissolved CA(HCO)3 in water (use particle illustration)
  - ARROW between them showing CO_2 + H_2O reaction

Text Content:

Header:
"How Cliffs Dissolve: The Weathering Reaction"

Chemical Equation (LaTeX-rendered, centered):
$CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$

Explanation (24pt, below equation):
"Rainwater + carbon dioxide = weak acid
Weak acid + chalk = soluble calcium bicarbonate
Result: ~1mm of cliff lost per year"

Visual Label:
Arrow annotation: "Natural erosion process"

Layout:

- Equation positioned in upper third
- Visual diagram in middle third
- Explanatory text in lower third
- Use color coding: Reactants (blue tones), Products (green tones)

SLIDE 4: The Kent Connection

Purpose: Ground the chemistry in local history and context.

Visual Elements:

- **Timeline Graphic:** Horizontal timeline from 100M years ago to present
    - Illustrated markers for key events:
        1. Cretaceous Period (microscopic algae illustration)
        2. Chalk formation (geological strata)
        3. Medieval lime burning (kiln illustration)
        4. Modern conservation (scientific monitoring)

Text Content:

Header:
"Kent's Cretaceous Chemistry"

Timeline Labels:
📍 100M BCE: Warm shallow sea covers Kent
📍 66M BCE: Chalk deposits reach 350ft depth
📍 1200s CE: Lime kilns power Canterbury's construction
📍 2025: Ocean acidification threatens marine calcite

Bottom Text:
"The same chemistry built our heritage—
and now faces climate pressure"

Layout:

- Timeline runs horizontally through center
- Header at top
- Bottom text in gold box for emphasis
- Icons/illustrations at each timeline point

SLIDE 5: The Call-to-Action

Purpose: Convert viewers into followers and direct traffic.

Visual Elements:

- **Logo Showcase:** CATS Insights logo enlarged (400×400px, centered)
- **QR Code:** Direct link to White Cliffs blog post on CATS Insights website (we need to make this still)
- Then an arrow and a call to see the caption where the article will be
    - Position: Bottom-left, 200×200px
    - Include instruction: "Scan for full article"
    - See caption for full article

Text Content:

Header:
"Explore More Chemistry"

Subheader:
"CATS Insights delivers high-level
science explained clearly"

Call-to-Action Buttons (stylized):
[FOLLOW] for more science deep dives
[VISIT] catsinsights website for full articles
[JOIN] our elective

Layout:

- Logo at top center
- Some incentives and why they should join cats insights
- QR code bottom-left
- buttons bottom-right as stacked list

Post Action Checklist

Instagram Caption Copy

🪨 THE CHEMISTRY OF KENT'S WHITE CLIFFS 🪨

Ever wondered why the White Cliffs are so... white? The answer lies 100 million years in the past.

💠 Calcium carbonate ($CaCO_3$) from ancient marine algae
💠 Trigonal crystal structure reflects pure white light
💠 Still eroding today through natural acid-rain chemistry

Swipe through to discover:
→ The molecular structure holding up the cliffs
→ The reaction slowly dissolving them
→ How medieval Kent turned chalk into quicklime
→ Why ocean acidification threatens this chemistry

🔬 This is chemistry that shaped our heritage—and still shapes our coastline.

📖 Read the full deep dive (link in bio)
🧪 Follow @catsinsights for science explained by students

#ChemistryExplained #WhiteCliffs #KentScience #STEMEducation #CalciumCarbonate #GeochemistryRocks #StudentResearch #SciComm #CATSInsights #ChemTwitter #RSCKent

Caption Metadata:

- Hashtag count: 11 (Instagram recommended max: 15)
- Target audience: A-Level students, teachers, science enthusiasts
- Engagement hooks: Question in first line, emoji bullets for scannability

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Analytics Tracking (Post-Publication)

Capture these metrics 72 hours after posting:

[ ] Total Reach (unique accounts)
[ ] Engagement Rate (likes + comments + shares / reach)
[ ] Profile Visits (traffic driven)
[ ] Swipe-Through Rate (% who viewed all 5 slides)
[ ] Link Clicks (if using link sticker in Stories repost)
[ ] Saves (indicates high-value content)

Screenshot Requirements for RSC Evidence:

- Instagram Insights overview page
- Detailed breakdown by slide (if available)
- Any notable comments/shares from science educators

Design Handoff Checklist

For Nana (Designer):

[ ] Confirm access to Canva/Adobe CC
[ ] Source high-res White Cliffs photo (CC0 license)
[ ] Generate $CaCO_3$ 3D model (ChemDraw/Jmol/Blender)
[ ] Create LaTeX equation PNGs (use codecogs.com or local LaTeX)
[ ] Apply CATS Insights brand guidelines (colors, fonts, logo)
[ ] Export 5 individual slides + master PSD/AI file
[ ] Deliver by Feb 16 for Jason's review
[ ] Final approval → Export for upload by Feb 18

Review Criteria (Jason's Approval):

- [ ] Scientific accuracy (equations, structures, facts)
- [ ] Visual hierarchy guides eye naturally
- [ ] Brand consistency across all 5 slides
- [ ] Text is readable on mobile screens
- [ ] No copyright violations (images, fonts)

Contingency: If Timeline Slips

Minimum Viable Carousel (3 slides instead of 5):

1. Hook (Slide 1)
2. The Molecule + Reaction combined (Slides 2+3 merged)
3. CTA (Slide 5)

Emergency Design Option:

- Use Canva's "Science Presentation" templates
- Manually input text from this brief
- Replace stock images with Chemistry-specific graphics

Would you like me to:

1. Generate the LaTeX equation graphics as downloadable PNGs?
2. Write the "Workshop Lesson Plan" (Activity 3 from your timeline)?
3. Create a project management Gantt chart with all deadlines?
4. Draft the "Award Alignment Document" for your teacher meeting?

Pick your next priority and I'll deliver immediately.

Action 5

Action 5: "Writing Chemistry for Non-Chemists" Workshop

Strategic Context

This workshop serves **three simultaneous objectives**:

1. **RSC Category 4 Evidence:** Proves you mentor student contributors (not just manage them)
2. **CATS Insights Operational Goal:** Trains your writers to meet publication standards
3. **Internal Credibility:** Demonstrates to your Head of Chemistry that you're building **sustainable systems**, not one-off projects

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Workshop Blueprint

Core Structure (45 minutes)

00:00-00:05 | Opening Hook: "The Translation Problem"
00:05-00:15 | Teaching Block 1: The 3-Level Framework
00:15-00:25 | Live Editing Exercise: White Cliffs Example
00:25-00:35 | Small Group Practice: "Fix This Paragraph"
00:35-00:42 | Debrief & Quality Standards Review
00:42-00:45 | Next Steps & Expectations

Section-by-Section Content

Opening Hook (5 minutes)

Your Script:

"Show of hands: Who here has written a science essay that your teacher said was 'too technical' or 'hard to follow'? [Pause for hands]

Now, different question: Who has read a science article online and thought, 'This is dumbed down—they're treating me like I'm stupid'? [Pause]

That's the translation problem. Chemistry has three audiences:

1. **Experts** (your chemistry teacher, university professors)

2. **Educated non-specialists** (A-Level students in *other* subjects)
3. **General public** (your parents, younger students)

Most of you write for Audience 1 because that's who grades your exams. But CATS Insights publishes for Audience 2. That's the skill we're learning today: **how to explain complex chemistry clearly without oversimplifying**.

The article I'm about to show you—on the White Cliffs—is written at exactly this level. Let's reverse-engineer how it works."

Teaching Block 1: The 3-Level Framework (10 minutes)

Visual Aid: Create a simple 3-column table (project on screen or print as handout)

Audience Level	$CaCO_3$ Explanation Example	When to Use
Level 1: Expert	"Calcite exhibits rhombohedral cleavage due to its trigonal crystal system (space group $R\bar{3}c$), with Ca^{2+} ions in 6-fold coordination with planar CO_3^{2-} groups."	Academic papers, university coursework
Level 2: Educated Non-Specialist	"Calcium carbonate forms a trigonal crystal structure where each calcium ion bonds with carbonate groups in a lattice that reflects light uniformly—creating the pure white color."	**CATS Insights target**
Level 3: General Public	"The cliffs are white because they're made of a mineral that bounces light back evenly, like a perfect mirror for all colors."	Science blogs, YouTube explainers

Your Narration:

"Notice what Level 2 does:

- ✓ Uses the technical term ('trigonal crystal structure') but explains its significance immediately
- ✓ Keeps one technical detail (lattice, ions) but cuts jargon ('rhombohedral cleavage', 'space group')
- ✓ Connects chemistry to observable reality ('reflects light uniformly → white color')

Level 1 assumes you already know what 'rhombohedral cleavage' means. Level 3 avoids all technical terms. Level 2 **teaches the technical terms in context**. That's our standard."

Live Editing Exercise (10 minutes)

Display the White Cliffs Article (Paragraph 3) on Screen:

Read aloud the weathering reaction paragraph:

> "The cliffs are not static. Weathering processes continuously reshape them through a reversible chemical reaction:
>
> $CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$
>
> Rainwater, slightly acidified by dissolved atmospheric carbon dioxide, converts solid calcite into soluble calcium bicarbonate..."

Analysis Questions (Ask the Room):

1. "What technical terms did I use?" [calcite, calcium bicarbonate, acidified]
2. "Did I define them explicitly?" [No—but context makes them clear]
3. "Could someone who's never taken A-Level Chemistry follow this?" [Yes, because you describe *what's happening* before naming it]

Key Teaching Point:

"Notice the structure:

1. **First**, tell them what's happening in plain language ('weathering reshapes the cliffs')
2. **Then**, show the chemical equation
3. **Finally**, walk through the mechanism step-by-step

This is the **PEEL structure** applied to science writing:

- **Point:** The cliffs are dynamic
- **Evidence:** The reaction equation
- **Explanation:** How rainwater + CO_2 creates acid
- **Link:** This demonstrates the carbon cycle

You all learned PEEL in English class. It works for chemistry too."

Small Group Practice (10 minutes)

The Task:

"I'm giving each group a paragraph from a *real* student draft. Your job: Identify whether it's written at Level 1, 2, or 3, then rewrite it for Level 2 (CATS Insights standard)."

Example Paragraph (Intentionally Level 1 - Too Technical):

> "The Haber-Bosch process operates via heterogeneous catalysis on an iron surface promoted by K_2O and Al_2O_3. The rate-determining step involves

> dissociative chemisorption of dinitrogen, overcoming the 945 kJ/mol bond enthalpy of the N≡N triple bond."

Expected Student Rewrite (Level 2):

> "The Haber-Bosch process uses an iron catalyst to break apart nitrogen molecules—a difficult task because the N≡N triple bond is exceptionally strong (945 kJ/mol). This bond-breaking step is the slowest part of the reaction, which is why the process requires high temperatures and pressure to produce ammonia at industrial scale."

Your Role During Group Work:

- Circulate between groups
- Ask guiding questions: "What's the most important idea in this paragraph?" "Which technical term is essential vs. which is jargon?"
- **This is when you should be recording video** (see Evidence Capture below)

Debrief & Quality Standards (7 minutes)

Reconvene the Room:

"Let's hear one rewrite from each group." [Call on 3 groups, max 1 minute each]

After They Share:

"Notice what you all did instinctively:

- You kept the important numbers (945 kJ/mol bond enthalpy)
- You explained *why* it matters (this is why the process is difficult)
- You cut the jargon that doesn't add information ('promoted by K_2O and Al_2O_3')

This is **editorial judgment**. It's not about dumbing down—it's about **clarity without loss of accuracy**.

Now, here's the CATS Insights standard: Every article you submit must pass this test:

1. **The Specialist Test:** Would a chemistry teacher correct any facts? [No]
2. **The Peer Test:** Could an A-Level History student follow the logic? [Yes]
3. **The Citation Test:** Are all claims backed by sources? [Yes]

If your draft fails any of these, the Lead Editor will send it back for revision. That's not punishment—it's mentorship. We're training you to write at university level."

Next Steps & Expectations (3 minutes)

Action Items for Attendees:

"By next week's meeting:

1. Choose a chemistry topic from the Topic Menu (handout provided)
2. Write a 300-word 'Level 2' explanation of one key concept from that topic
3. Submit via the Google Form (link in the group chat)

Your draft will go through our editorial process:

- **Round 1:** Peer review by another writer
- **Round 2:** Feedback from Lead Editor (Grisha)
- **Round 3:** My final approval

The goal is **not** to get it perfect on the first try. The goal is to **improve measurably** from Draft 1 to Draft 3. That's what universities look for: **intellectual growth**.

Questions?"

Evidence Capture Plan (RSC Category 4 Proof)

During the Workshop:

Video Recording (2-3 minutes total):

[ ] 30 seconds: You explaining the 3-Level Framework (capture the teaching moment)
[ ] 60 seconds: Pan across groups during the Small Group Practice (shows engagement)
[ ] 30 seconds: A student sharing their rewrite during debrief (shows learning outcome)
[ ] 30 seconds: You delivering the "Next Steps" (shows structured follow-through)

Photography (5-10 shots):

[ ] Wide shot of the room (shows workshop setup)
[ ] Close-up of handout materials (shows preparation)
[ ] Over-the-shoulder shot of a student's rewrite (anonymize if needed)
[ ] Group selfie at the end (shows team cohesion)

Assign a Photographer: Duc or Cindy (they're your secretaries—this is administrative documentation)

After the Workshop:

Collect Before/After Writing Samples (Within 7 Days):

1. **Before:** The "Level 1" paragraph you gave them in the exercise
2. **After:** Their submitted 300-word draft (due Feb 26)

For RSC Evidence, You Need:

- 2-3 anonymized examples showing improvement
- Specific metrics: "Student A reduced jargon by 60% while maintaining factual accuracy"
- Quotes from students: "I didn't realize I was writing for experts only—this helped me think about my reader"

Workshop Impact Report Template

Create a 1-page document (due Feb 27) with this structure:

CATS Insights Workshop: "Writing Chemistry for Non-Chemists"

Date: February 19, 2026 | **Duration:** 45 minutes | **Facilitator:** Jason Ye Yint

Attendees: [Number] CATS Insights members (Years 9-12)

Learning Objectives:

1. Distinguish between expert, educated non-specialist, and general public communication levels
2. Apply PEEL structure to scientific explanation
3. Practice editorial judgment in simplifying without oversimplifying

Methodology:

- 10-minute teaching block (3-Level Framework)
- 10-minute live editing demonstration (White Cliffs article analysis)
- 10-minute small group practice (paragraph rewriting exercise)
- 7-minute debrief and quality standards review

Quantitative Outcomes:

- **Pre-Workshop:** [X]% of participants could not explain why their science essays were marked "unclear"
- **Post-Workshop:** [Y]% successfully rewrote a technical paragraph to Level 2 standard
- **Follow-Up:** [Z] students submitted drafts within 7 days (compliance rate)

Qualitative Feedback:

- "I always thought 'simplifying' meant removing accuracy—now I see it's about structuring information better." —Student B
- "The PEEL framework finally makes sense outside of English class." —Student F

Measurable Improvement (Sample):

- **Student C Draft 1:** 12 undefined technical terms, 2 sentences >40 words
- **Student C Draft 3:** 3 technical terms (all defined in context), average sentence length 18 words, clarity score +65%

Next Steps: All attendees are now entering the editorial workflow. Expected publication-ready articles: [Date]

Delegation Instructions (Immediate Actions)

To Cindy (Head of Academic Development):

"I need you to co-facilitate this workshop with me. Your role:

1. Print 15 copies of the handout (3-Level Framework table + Topic Menu)
2. Bring your phone to record 2-3 minutes of video during the group exercise
3. Take the attendance list—we need names for the Impact Report
4. After the workshop, follow up with students who don't submit drafts within 7 days

Deadline: Workshop runs Week of Feb 19. I need the handouts printed by Feb 18."

To Grisha (Lead Editor):

"After this workshop, you'll receive 15-20 draft submissions. Your job:

1. Use the 3-Level Framework as your review criteria
2. Track how many revisions each draft needs before it's publication-ready
3. Document specific improvements (e.g., 'Student reduced jargon by X%, improved clarity score by Y%')

This data becomes our RSC Award evidence for mentorship impact.

Deadline: All drafts reviewed by March 1."

To Duc (Secretary):

"Create a Google Form for workshop attendees to submit their 300-word drafts. Fields:

- Name
- Year Group
- Topic chosen from Topic Menu
- Draft text (paragraph box)
- One question they have about their draft

Share the form link at the end of the workshop.

Deadline: Form created by Feb 18."

Workshop Logistics Checklist

Venue: Knights Exam Room (your preferred space—35 student capacity)

Date/Time Options:

Option A: Thursday Feb 20, 4:05-5:00 PM (regular meeting time)
Option B: Saturday Feb 22, 2:00-3:00 PM (weekend session for those with conflicts)

Materials Needed:

[ ] Laptop + projector (to display White Cliffs article)
[ ] Printed handouts (15 copies of 3-Level Framework + Topic Menu)
[ ] Whiteboard + markers (for live note-taking during debrief)
[ ] Phone/camera for video recording
[ ] Attendance sheet
[ ] Google Form link (display as QR code at the end)

Promotion (Announce at Feb 13 Meeting):

"Next week, we're running a workshop on how to write chemistry clearly. This is **mandatory for all writers** and **recommended for editors**. If you want your articles to be published, you need to understand the CATS Insights communication standard. See you Thursday 4:05 PM in Knights."

Contingency Plans

If Attendance is Low (<10 students):

Pivot Strategy: Run it as a "Masterclass" format

- Smaller group = more individualized feedback

- Record the entire session (15-20 minutes) for future onboarding
- Position it as "exclusive training for the founding cohort"

If Students Struggle with the Exercise:

Backup Activity: Show them two versions of the same paragraph (one Level 1, one Level 2) and ask them to identify the differences. This reduces the cognitive load from "create" to "analyze."

If Recording Fails:

Evidence Backup: Take detailed notes during the workshop, then do a 2-minute "video summary" afterward where you explain what was taught and show student work samples on camera.

Success Metrics

You'll know the workshop succeeded if:

1. **Immediate:** Students submit drafts within 7 days (target: 80%+ compliance)
2. **Medium-term:** Drafts require fewer revision rounds than pre-workshop articles (track this quantitatively)
3. **Long-term:** RSC judges watch the workshop video and say, "This student is teaching university-level communication skills"

Your Next Decision Point

After this workshop, you'll have 3 parallel workstreams running:

1. **Action 4 finalization:** Nana completes Instagram carousel (Feb 18 deadline)
2. **Action 5 execution:** You run the workshop + collect evidence (Feb 19-27)
3. **Action 6 prep:** You begin drafting the "Teaching Resource" infographic (Week 5: Feb 26-Mar 4)

Can your team handle 3 concurrent projects? If yes, I'll brief you on Action 6 prep now. If no, we pause and consolidate.

What's your call?

Step by step pla

Now I have comprehensive context. Let me build you the complete chronological execution plan for Action 5.

ACTION 5: MASTER EXECUTION PLAN

"Writing Chemistry for Non-Chemists" Workshop

PHASE 1: PRE-WORKSHOP PREPARATION

(T-7 to T-1 Days)

STEP 1: Materials Creation *(T-7 Days)*

Task	Owner	Deliverable	Status Check
1.1	**You**	Finalize the **3-Level Framework Handout** (the 3-column table showing Expert/Educated Non-Specialist/General Public examples)	☐ Saved as PDF
1.2	**You**	Create the **"Fix This Paragraph" Exercise Sheet** with the Level 1 paragraph from the White Cliffs article that students will rewrite	☐ Printed x25 copies
1.3	**You**	Prepare the **Topic Menu Handout** listing 8-10 chemistry topics students can choose from for their 300-word homework	☐ Printed x25 copies
1.4	**You**	Build the **Presentation Slides** covering the 45-minute structure	☐ Tested on projector
1.5	**Cindy** (Head of Academic Development)	Set up the **Google Form** for homework submission (due Feb 26)	☐ Link tested

Materials Checklist:

[ ] 3-Level Framework Handout (25 copies)
[ ] "Fix This Paragraph" Exercise Sheet (25 copies)
[ ] Topic Menu (25 copies)

[ ] Attendance Sheet (1 copy)
[ ] Feedback Form - optional (25 copies)
[ ] Pens/Pencils for students who forget

STEP 2: Logistics Confirmation *(T-5 Days)*

Task	Owner	Action
2.1	**You**	Book the room with Miss Emma (confirm projector, whiteboard access)
2.2	**You**	Confirm workshop date/time in Teams General channel
2.3	**Secretary (Duc/Elain)**	Update the CATS Insights shared calendar

---

STEP 3: Announcement & Marketing *(T-4 Days)*

3.1 — Teams Announcement (Post this verbatim):

> 📢 **MANDATORY WORKSHOP: "Writing Chemistry for Non-Chemists"**
>
> **When:** [DATE], [TIME] **Where:** [ROOM] **Duration:** 45 minutes
>
> **What you'll learn:** ✓ How to explain chemistry without dumbing it down ✓ The "Level 2 Writing" framework used by professional science journalists ✓ PEEL structure for clear scientific paragraphs
>
> **What to bring:** Pen/pencil, your brain.
>
> **Homework assigned at the end:** A 300-word draft due Feb 26. This counts toward your CATS Insights contribution.
>
> **Attendance is mandatory.** React with ✅ to confirm you've read this.

3.2 — Pre-Commitment Survey (Optional but Recommended):

Create a 2-question Microsoft Form:

- "Will you attend on [DATE]?" (Yes/No/Conflict—please explain)
- "What chemistry topic interests you most?" (Open text)

This gives you attendance forecast + topic preferences for the Topic Menu.

STEP 4: Evidence Team Briefing *(T-2 Days)*

4.1 — Assign Photographer/Videographer: Delegate to **Duc** or **Aung** (Secretaries). Brief them on exactly what to capture:

Video Recording Checklist (2-3 minutes total):

[ ] 30 sec: You explaining the 3-Level Framework (capture the teaching moment)
[ ] 60 sec: Pan across groups during Small Group Practice (shows engagement)
[ ] 30 sec: A student sharing their rewrite during debrief (shows learning outcome)
[ ] 30 sec: You delivering "Next Steps" (shows structured follow-through)

Photography Checklist (5-10 shots):

[ ] Wide shot of room setup (BEFORE students arrive)
[ ] Close-up of handout materials spread out
[ ] You presenting at the front
[ ] Students working in groups
[ ] Over-the-shoulder shot of a student's rewrite (anonymize faces if needed)
[ ] Group photo at the end

4.2 — Equipment Check:

[ ] Photographer's phone charged to 100%
[ ] Backup device identified
[ ] Storage space cleared (minimum 2GB free)
[ ] Test video/photo quality in the room lighting

STEP 5: Final Prep *(T-1 Day / Night Before)*

Task	Check
Print all materials	☐
Test presentation on your laptop	☐
Charge laptop to 100%	☐
Pack materials in one folder/bag	☐
Confirm with photographer via DM	☐
Set phone alarm for 30 min before workshop	☐
Re-read your opening script (memorize the hook)	☐

PHASE 2: WORKSHOP DAY EXECUTION

(T-Day)

STEP 6: Room Setup *(30 minutes before)*

Task	Time
Arrive early, arrange desks for group work (clusters of 4-5)	T-30
Test projector/screen	T-25
Place handouts on each desk	T-20
Write key terms on whiteboard: **"Level 1 / Level 2 / Level 3"**	T-15
Photographer takes "room setup" shot	T-10
Position photographer at back-corner with clear sightline	T-5

STEP 7: Attendance & Opening *(00:00–00:05)*

Action	Detail
Secretary (Duc/Elain) circulates attendance sheet	Capture full names for RSC evidence
Photographer begins recording your opening hook	Capture the "Translation Problem" segment
Deliver the **Opening Hook** script	Use the exact language from the blueprint—it's optimized

Opening Hook Script (Memorize This):

> "Show of hands: Who here has written a science essay that your teacher said was 'too technical' or 'hard to follow'? [Pause]
>
> Different question: Who has read a science article online and thought, 'This is dumbed down—they're treating me like I'm stupid'? [Pause]
>
> That's the translation problem. Chemistry has three audiences: Experts, Educated non-specialists, and the General public. Most of you write for Audience 1 because that's who grades your exams. But CATS Insights publishes for Audience 2. That's the skill we're learning today."

STEP 8: Teaching Block 1 *(00:05–00:15)*

Action	Detail
Project the 3-Level Framework table	Walk through each row
Ask students to identify which level the White Cliffs blog uses	Answer: Level 2
Photographer captures you teaching	30-second clip

Teaching Points to Hit:

1. Level 1 (Expert) = crystallography jargon, assumes prior knowledge
2. Level 2 (Educated Non-Specialist) = technical but explained, assumes curiosity
3. Level 3 (General Public) = oversimplified, "dumbed down"

STEP 9: Live Editing Exercise *(00:15–00:25)*

Action	Detail
Show a "before" paragraph (Level 1 jargon-heavy) on screen	Pull from White Cliffs draft
Model how YOU would rewrite it to Level 2	Think aloud, show your reasoning
Highlight the PEEL structure in your rewrite	Point—Evidence—Explanation—Link

Key Teaching Move: Explicitly say "Watch my process" before you edit. This frames you as a mentor, which is RSC Category 4 evidence.

STEP 10: Small Group Practice *(00:25–00:35)*

Action	Detail
Distribute the "Fix This Paragraph" exercise sheet	Each group gets the same Level 1 paragraph
Groups of 4-5 rewrite it to Level 2	Circulate and coach
Photographer pans across groups working	60-second video

Circulating Prompts (Use These When Coaching Groups):

- "Would your History A-Level friend understand this?"

- "You've kept the fact—now explain *why* it matters."
- "Good—you removed jargon. But did you replace it with clarity or vagueness?"

STEP 11: Debrief & Quality Standards *(00:35–00:42)*

Action	Detail
Call on 2-3 groups to share their rewrites	Ask them to read aloud
Apply the **Three-Test Framework** publicly	1. Specialist Test, 2. Peer Test, 3. Citation Test
Photographer captures a student sharing	30-second clip

Three-Test Framework (Say This):

> "Before you submit anything to CATS Insights, run these three tests:
>
> 1. **The Specialist Test:** Would a chemistry teacher correct any facts?
> 2. **The Peer Test:** Could an A-Level History student follow the logic?
> 3. **The Citation Test:** Are all claims backed by sources?"

STEP 12: Next Steps & Close *(00:42–00:45)*

Action	Detail
Distribute the **Topic Menu** handout	Students choose their homework topic
Explain the homework assignment	300 words, Level 2 writing, due Feb 26
Share the Google Form submission link	Post in Teams chat immediately
Photographer captures your closing	30-second clip

Closing Script:

> "By next week: Choose a topic from the menu. Write a 300-word Level 2 explanation. Submit via the Google Form.
>
> Your draft will go through our editorial process—peer review, Lead Editor feedback, then my final approval. The goal is NOT to get it perfect on Day 1. The goal is to improve measurably from Draft 1 to Draft 3. That's what universities look for: intellectual growth.
>
> Questions?"

STEP 13: Post-Workshop Evidence Capture *(Immediately After)*

Task	Owner	Deadline
Group photo with all attendees	Photographer	T+5 min
Collect attendance sheet from secretary	You	T+5 min
Back up all video/photo to Google Drive	Photographer	T+1 hour
Post attendance count in Teams (e.g., "18/20 attended")	Secretary	T+1 hour

---

PHASE 3: POST-WORKSHOP FOLLOW-UP

(T+1 to T+14 Days)

STEP 14: Evidence Organization *(T+1 Day)*

Create a folder in Google Drive: **"RSC Evidence > Action 5 - Workshop"**

Folder Structure:

```
Action 5 - Workshop/
├── 01_Attendance/
│   └── Attendance_Sheet_Feb19.pdf
├── 02_Materials/
│   ├── 3_Level_Framework_Handout.pdf
│   ├── Fix_This_Paragraph_Exercise.pdf
│   └── Topic_Menu.pdf
├── 03_Media/
│   ├── Video_Opening_Hook.mp4
│   ├── Video_Group_Work.mp4
│   ├── Video_Debrief.mp4
│   ├── Video_Closing.mp4
│   └── Photos/
├── 04_Student_Work/
│   ├── Before_Samples/
│   └── After_Samples/
└── 05_Impact_Report/
    └── Workshop_Impact_Report.pdf
```

STEP 15: Collect Homework Submissions *(T+7 Days / Feb 26)*

Task	Owner
Export all Google Form submissions	Cindy
Compile into a single folder	Cindy
Flag top 3 for RSC evidence (best improvement)	Grisha (Lead Editor)

STEP 16: Generate Before/After Comparisons *(T+8 Days)*

For RSC Evidence, You Need:

- 2-3 anonymized student examples showing improvement
- Specific metrics (e.g., "Student A reduced jargon by 60% while maintaining factual accuracy")
- Optional: A quote from a student ("I didn't realize I was writing for experts only—this helped me think about my reader")

How to Document Improvement:

Metric	Before (Level 1)	After (Level 2)
Jargon terms (unexplained)	8	2
Explanatory sentences	1	5
PEEL structure adherence	0%	80%
Word count	120	310

STEP 17: Write the Workshop Impact Report *(T+9 Days / Feb 27)*

Template (1-page document):

CATS Insights Workshop: "Writing Chemistry for Non-Chemists"

Date: [DATE] | **Duration:** 45 minutes | **Facilitator:** Jason Ye Yint

Attendees: [X] CATS Insights members (Years 9-12)

Learning Objectives:

1. Distinguish between expert, educated non-specialist, and general public communication levels
2. Apply PEEL structure to scientific explanation
3. Self-assess writing using the Three-Test Framework

Workshop Activities:

- 3-Level Framework lecture with White Cliffs case study
- Live editing demonstration
- Small group rewriting exercise
- Peer debrief and quality standards review

Evidence of Impact:

- [X]% of attendees submitted 300-word drafts by deadline
- Average improvement in jargon reduction: [X]%
- Student quote: "[Insert quote]"

Alignment with RSC Category 4 (Mentoring): This workshop demonstrates structured mentorship of student contributors. Participants received direct instruction in science communication methodology, practiced skills through guided exercises, and will continue to receive feedback through the CATS Insights editorial pipeline.

Attached Evidence:

- Attendance sheet
- Workshop materials (3 handouts)
- Video clips (4 segments, 2:30 total)
- Before/after writing samples (3 anonymized)

STEP 18: Final Evidence Package *(T+10 Days)*

Compile everything into a single submission-ready folder:

```
RSC_Action5_Evidence_Package/
├── Impact_Report.pdf
├── Attendance_Sheet.pdf
├── Media_Clips.zip
├── Before_After_Samples.pdf
└── Workshop_Materials.zip
```

CRITICAL SUCCESS METRICS

Metric	Target	How to Verify
Attendance	≥80% of CATS Insights members	Attendance sheet
Video evidence	≥2 minutes, all 4 segments captured	Media folder
Homework submission rate	≥70% of attendees	Google Form count
Documented improvement	≥2 before/after comparisons	Impact Report
Impact Report completed	By Feb 27	File timestamp

DELEGATION SUMMARY

Role	Person	Responsibilities
Workshop Lead	You	Deliver all content, approve final materials
Attendance & Admin	Duc or Elain	Circulate attendance, post to Teams
Photographer/Videographer	Duc or Aung	Capture all media per checklist
Google Form Setup	Cindy	Create submission form, export data
Homework Review	Grisha	Flag top 3 submissions for RSC evidence
Folder Organization	You (or delegate to Cindy)	Maintain Drive structure

---

NEXT IMMEDIATE ACTION

Today: Draft the 3-Level Framework Handout and the "Fix This Paragraph" exercise sheet. These are the two materials that cannot be delegated because they require your pedagogical judgment. Send me what you produce—I'll audit them for RSC alignment before you print.

3 level framwork

CATS INSIGHTS | Workshop Handout

The 3-Level Framework

Writing Chemistry for Non-Chemists

The Core Problem: Most students write chemistry for their teachers (experts). But CATS Insights publishes for ***educated non-specialists*** — intelligent readers who lack chemistry training. This framework teaches you how to hit that sweet spot.

Level	Example: Calcium Carbonate	Characteristics
LEVEL 1 **Expert** *(Too Technical)*	*"Calcite exhibits rhombohedral cleavage due to its trigonal crystal system (space group $R\bar{3}c$), with Ca^{2+} ions in 6-fold coordination with planar CO_3^{2-} groups."*	• Assumes prior knowledge • Unexplained jargon • Written for chemistry teachers • Exam-style writing
LEVEL 2 **Educated Non-Specialist** **★ TARGET ★**	*"Calcium carbonate adopts a trigonal crystal system—imagine the atoms arranged in a three-dimensional lattice where each calcium ion connects to surrounding carbonate groups. This regular structure reflects light uniformly, producing chalk's characteristic white colour."*	• Technical terms explained • Assumes curiosity, not expertise • Written for A-Level History students • CATS Insights standard
LEVEL 3 **General Public** *(Oversimplified)*	*"Chalk is made of tiny sea creatures! When they died millions of years ago, their shells piled up and turned into the white cliffs we see today."*	• Avoids all technical terms • Treats reader as unintelligent • Written for primary school • Sacrifices accuracy for simplicity

The Three-Test Framework

Before submitting to CATS Insights, run these three tests on your draft:

1. THE SPECIALIST TEST	2. THE PEER TEST	3. THE CITATION TEST
Would a chemistry teacher correct any facts? *(If yes → fix accuracy)*	Could an A-Level History student follow the logic? *(If no → add explanation)*	Are all claims backed by sources? *(If no → add references)*

Key Techniques for Level 2 Writing

Define on first use	*"coccolithophores—microscopic algae that build calcite shells"*
Use analogies	*"imagine the atoms arranged in a three-dimensional lattice" (visualisation aid)*
Connect to real world	*"This same property makes chalk soft enough to write with" (familiar application)*
Explain the 'so what'	*"producing chalk's characteristic white colour" (why does this matter?)*

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Topic menuy

CATS INSIGHTS | Homework Assignment

Topic Menu

Choose ONE topic for your 300-word Level 2 draft

ASSIGNMENT: Write a 300-word explanation of ONE chemistry concept from the topics below. Write at Level 2 (educated non-specialist). Submit via the Google Form by **February 26, 2026**.

🏴‍☠️ Kent Chemistry (Local Focus — RSC Award Relevant)

1. The Chemistry of Kentish Hops

Focus: Alpha acids (humulone) in hops and their isomerisation during brewing. Why does Kent produce such distinctive flavours?
Hook: *Kent was once called the "Garden of England" partly because of its hop fields.*

2. Pfizer's Sandwich Legacy: Sildenafil (Viagra)

Focus: How PDE5 inhibitors work at the molecular level. A drug discovered in Kent that changed medicine.
Hook: *One of the world's most famous drugs was developed just 25 miles from Canterbury.*

3. The Iron in Tunbridge Wells Spa Water

Focus: Chalybeate springs—how iron(II) compounds dissolve in groundwater and why people believed it was medicinal.
Hook: *In 1606, a spring was discovered that turned Tunbridge Wells into a health resort.*

🧪 Everyday Chemistry

4. Why Does Cutting Onions Make You Cry?

Focus: Syn-propanethial-S-oxide: the volatile sulfur compound released when onion cells are damaged. Enzyme-substrate chemistry in action.

5. The Maillard Reaction: Why Cooked Food Tastes Better

Focus: Amino acid-sugar reactions that create browning, flavour, and aroma. From toast to steak.

6. How Do Glow Sticks Work?

Focus: Chemiluminescence: the diphenyl oxalate-hydrogen peroxide reaction and energy transfer to fluorescent dyes.

🌿 Green Chemistry & Environment

7. Ocean Acidification: The Other CO_2 Problem

Focus: How dissolved CO_2 forms carbonic acid and threatens calcifying marine organisms. The chemistry behind coral bleaching.

8. Biodegradable Plastics: PLA vs Traditional Polymers

Focus: Polylactic acid (PLA) from corn starch vs petroleum-based polyethylene. Why "biodegradable" is more complicated than it sounds.

🔬 Advanced Chemistry (Challenge Topics)

9. CRISPR-Cas9: The Chemistry of Gene Editing

Focus: How the Cas9 enzyme recognises and cuts DNA at specific sequences. Guide RNA and the PAM sequence.

10. Lithium-Ion Batteries: Why Your Phone Dies in the Cold

Focus: Li^+ ion intercalation, electrolyte viscosity, and temperature-dependent kinetics. The chemistry inside every smartphone.

MY CHOSEN TOPIC: # ____ _____

Why I chose this: _____

Submit via Google Form by February 26 | Questions? Message Jason or Grisha

Fix the paragraph

CATS INSIGHTS | Workshop Exercise

Fix This Paragraph

Rewrite Level 1 → Level 2

YOUR TASK:

The paragraph below is written at Level 1 (too technical). Your group has 10 minutes to rewrite it at Level 2 (educated non-specialist). Remember: your reader is an intelligent A-Level History student who has never studied chemistry.

The Original Paragraph (Level 1 — Too Technical)

The White Cliffs of Dover undergo continuous chemical weathering via a reversible acid-base reaction. Atmospheric CO_2 dissolves in precipitation to form carbonic acid (H_2CO_3), which reacts with the calcite ($CaCO_3$) matrix via the equilibrium: $CaCO_3(s) + H_2O(l) + CO_2(g) \rightleftharpoons Ca(HCO_3)_2(aq)$. The resulting calcium bicarbonate solution percolates through the permeable chalk strata, causing dissolution along bedding planes and producing the characteristic vertical joint patterns. This represents the lithospheric component of the global carbon cycle, with calcium ions eventually returning to the marine environment where biogenic calcification by coccolithophores recycles the carbon into new $CaCO_3$ tests.

What Makes This Level 1?

✗ Unexplained jargon:
- "reversible acid-base reaction"
- "equilibrium"
- "permeable chalk strata"
- "lithospheric component"
- "biogenic calcification"
- "$CaCO_3$ tests"

✗ Missing context:
- Why should the reader care?
- What do the cliffs look like?
- How fast does this happen?
- What are coccolithophores?

✗ Assumes chemistry A-Level

Your Level 2 Rewrite

Before You Share: Quick Checklist

- ☐ Did you explain every technical term on first use?
- ☐ Could a History student follow your logic?
- ☐ Did you keep all the facts accurate? (No oversimplification)
- ☐ Did you connect the chemistry to something the reader can visualise?

Group Members: _____

Presentation Plan

CATS INSIGHTS

Workshop Presentation Plan

"Writing Chemistry for Non-Chemists"

Duration: 45 Minutes | 12 Slides

Presenter: Jason Ye Yint

DESIGN SPECIFICATIONS

Instructions for the Design Team

Colour Palette

PRIMARY	ACCENT	LIGHT	BACKGROUND
Navy Blue	Worthgate Gold	Ice Blue	White
#003366	#C9A227	#E6EEF5	#FFFFFF

Level-Specific Colours (for the 3-Level Framework)

LEVEL 1 (Expert)	LEVEL 2 (Target)	LEVEL 3 (General)
Red: #C62828 / BG: #FFCDD2	Green: #2E7D32 / BG: #C8E6C9	Orange: #E65100 / BG: #FFE0B2

Typography

Element	Font	Size
Slide Title	Georgia Bold	40-44pt
Subheading	Calibri Bold	24-28pt
Body Text	Calibri	18-20pt
Quotes/Examples	Calibri Italic	16-18pt

Layout Rules

- Aspect Ratio: 16:9 (widescreen)
- Margins: 0.5 inch minimum from all edges
- Footer: Include slide number (bottom right) and "CATS Insights" (bottom left)
- Visual Motif: Use rounded rectangles for content boxes; thin gold accent lines to separate sections
- Icons: Use simple line icons in blue; place in small circles where appropriate
- **NO underlines beneath titles (this looks AI-generated)**

Image Requirements

- Slide 1: White Cliffs of Dover photograph (royalty-free, dramatic angle)
- Slide 3: Molecular diagram or abstract chemistry visual
- Slide 8: Screenshot mockup or document icon
- All other slides: Use icons or geometric shapes rather than photographs

SLIDE-BY-SLIDE BREAKDOWN

SLIDE 1: Title Slide [00:00-00:30]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
CATS INSIGHTS "Writing Chemistry for Non-Chemists" Workshop | [DATE] Presented by Jason Ye Yint	*No spoken script. Walk to the front, wait for silence, click to the next slide.*	LAYOUT: Full-bleed image of White Cliffs on right 60%; text on left 40% with dark blue overlay Title in white, large (48pt) Subtitle in gold, smaller (28pt) CATS Insights logo in top-left corner Dark/dramatic feel to set professional tone

SLIDE 2: The Translation Problem [00:30-01:30]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
"Show of hands..." Two questions to audience: 1. "Too technical" feedback? 2. "Dumbed down" article online? [Empty space for visual]	*"Show of hands: Who here has written a science essay that your teacher said was 'too technical' or 'hard to follow'? [PAUSE] Now, different question: Who has read a science article online and thought, 'This is dumbed down—they're treating me like I'm stupid'? [PAUSE] That's the translation problem."*	LAYOUT: Two-column with icons Left: Red warning icon + "Too Technical" Right: Orange caution icon + "Too Simple" Central question mark graphic between them Minimal text—this slide is for engagement, not reading

SLIDE 3: The Three Audiences [01:30-02:30]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
Chemistry has THREE audiences: 1. EXPERTS (Your chemistry teacher) 2. EDUCATED NON-SPECIALISTS (A-Level History students) 3. GENERAL PUBLIC (Your parents)	*"Chemistry has three audiences: [click] Experts—your chemistry teacher, university professors. [click] Educated non-specialists—A-Level students in OTHER subjects, like History. [click] General public—your parents, younger students. Most of you write for Audience 1 because that's who grades your exams."*	LAYOUT: Three horizontal cards/boxes stacked vertically Card 1 (top): Red background, mortar board icon Card 2 (middle): GREEN background (largest), book icon — highlight this as TARGET Card 3 (bottom): Orange background, family icon Use reveal animation (appear one at a time)

SLIDE 4: CATS Insights Target [02:30-03:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES

CATS Insights publishes for: LEVEL 2: Educated Non-Specialists "Explain complex chemistry clearly without oversimplifying."	*"But CATS Insights publishes for Audience 1 and 2. But we all already know how to present for Experts in the field but not for educated non-specialists.That's the skill we're learning today: how to explain complex chemistry clearly without oversimplifying. The article I'm about to show you—on the White Cliffs—is written at exactly this level. Let's reverse-engineer how it works."*	LAYOUT: Large central text with green highlight box Big "LEVEL 2" in green Tagline in italics below Simple, clean, impactful — this is the "thesis" slide Maybe a subtle chemistry molecule graphic in corner

SLIDE 5: The 3-Level Framework [03:00-08:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
[TABLE: 3 columns x 3 rows] Headers: Level | Example ($CaCO_3$) | When to Use Row 1: LEVEL 1 Expert | "Calcite exhibits rhombohedral cleavage..." | Exam essays Row 2: LEVEL 2 Target | "Calcium carbonate adopts a trigonal crystal system—imagine..." | CATS Insights Row 3: LEVEL 3 General | "Chalk is made of tiny sea creatures!" | Primary school	*"Here's the framework. [Walk through each row] Level 1 is what you write in exams—assumes your reader already knows the jargon. Level 2 is our target—technical but explained. Level 3 is oversimplified—sacrifices accuracy for ease. Notice how Level 2 keeps the word 'trigonal' but immediately explains what it means."*	LAYOUT: Full-width table Use colour-coded rows: Red for Level 1, Green for Level 2, Orange for Level 3 Level 2 row should be visually largest/most prominent Keep example text readable but not too long This is the KEY TEACHING SLIDE — spend time here

SLIDE 6: The White Cliffs Example [08:00-10:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
CASE STUDY: White Cliffs of Dover Original text excerpt: "The cliffs owe their distinctive snowy chalky white colour to calcium carbonate ($CaCO_3$), specifically the mineral calcite." Why is this Level 2? • Names the compound (technical) • But explains what calcite IS • Connects to what you can SEE	*"Look at this sentence from the White Cliffs article. [Read excerpt] Why is this Level 2? It names the compound—that's technical. But it explains that calcite IS calcium carbonate. And it connects to something visual: the snowy white colour you can actually see."*	LAYOUT: Two-column Left: Photo of White Cliffs Right: Quote box with excerpt, then bullet points below Highlight key phrases in the excerpt ("calcium carbonate", "calcite") with underline or bold Green checkmark icons next to bullet points

SLIDE 7: Key Techniques [10:00-13:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
HOW to write Level 2: 1. DEFINE ON FIRST USE "coccolithophores—microscopic algae" 2. USE ANALOGIES "imagine atoms arranged in a lattice" 3. CONNECT TO REAL WORLD "soft enough to write with" 4. EXPLAIN THE 'SO WHAT' "producing chalk's white colour"	*"Four techniques. [Go through each with examples from the article] One: Define on first use—when I write coccolithophores, I immediately say what they are. Two: Use analogies—'imagine the atoms arranged in a lattice.' Three: Connect to the real world—'soft enough to write with.' Four: Explain the so what—why does crystal structure matter? Because it produces the white colour."*	LAYOUT: Four horizontal cards with icons Each card: Icon on left, technique name bold, example in italics below Icons: Dictionary, lightbulb, globe, question mark Consistent styling across all four Consider numbered circles (1-4) for easy reference

SLIDE 8: Exercise Instructions [13:00-15:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
EXERCISE: Fix This Paragraph You have: 10 MINUTES Task: • Read the Level 1 paragraph • Rewrite it at Level 2 • Use the techniques we just covered Work in groups of 4-5	*"Now it's your turn. You have 10 minutes. On your handout, there's a Level 1 paragraph from the White Cliffs article. Your job: rewrite it at Level 2. Use the four techniques. Work with the people at your table. I'll walk around."*	LAYOUT: Large timer/clock icon "10 MINUTES" in large bold text Task list in clear bullet points Simple and scannable—students need to understand quickly and then look at their handout, not the screen

SLIDE 9: Timer / Circulate Slide [15:00-25:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
[LARGE COUNTDOWN TIMER] 10:00 Remember: • Would a History student understand this? • Are facts still accurate?	*[NO SCRIPT — circulate and coach groups. Use these prompts: "Would your History A-Level friend understand this?" / "You've kept the fact—now explain WHY it matters." / "Good—you removed jargon. But did you replace it with clarity or vagueness?"]*	LAYOUT: Full-screen timer (use Canva/PP built-in countdown or just large numbers) Two reminder bullet points at bottom in smaller text Minimal distraction—students should focus on their handout

SLIDE 10: The Three Tests [25:00-32:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
Before submitting to CATS Insights: 1. THE SPECIALIST TEST Would a chemistry teacher correct any facts? (If yes → fix accuracy) 2. THE PEER TEST Could a History student follow the logic? (If no → add explanation) 3. THE CITATION TEST Are all claims backed by sources? (If no → add references)	*"Before anything goes on CATS Insights, it passes three tests. [Click through each] The Specialist Test: Would a chemistry teacher correct any facts? The Peer Test: Could an A-Level History student follow the logic? The Citation Test: Are all claims backed by sources? If your draft fails any of these, the Lead Editor sends it back. That's not punishment—it's mentorship."*	LAYOUT: Three large numbered boxes/cards Each test gets its own prominent space Use icons: checkmark, peer group, book/citation Colour-code: Blue for all three (neutral, quality-focused) This is the QUALITY STANDARDS slide

SLIDE 11: Your Homework [32:00-40:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
BY NEXT WEEK: 1. Choose a topic from the Menu 2. Write 300 words (Level 2) 3. Submit via Google Form DEADLINE: February 26 https://docs.google.com/forms/d/e/1FAIpQLSdWIUG9O0kQ6ipIznOmcHx5x0YEqST5H6vmAILC6EcOD4mJ_A/viewform?usp=header	*"By next week's meeting: [Click] Choose a chemistry topic from the Topic Menu handout. [Click] Write a 300-word Level 2 explanation of one key concept. [Click] Submit via the Google Form—link is in the Teams chat and there's a QR code here. Your draft will go through our editorial process: peer review, Lead Editor feedback, then my final approval."*	LAYOUT: Numbered steps on left, QR code on right Large, clear QR code that scans easily from back of room Deadline in red or gold to stand out Keep simple—students need to remember three things

SLIDE 12: Closing / Q&A [40:00-45:00]

SLIDE CONTENT	SPEAKER SCRIPT	DESIGN NOTES
The goal is NOT perfection. The goal is MEASURABLE IMPROVEMENT. Draft 1 → Draft 3 That's what universities look for: INTELLECTUAL GROWTH Questions?	*"The goal is NOT to get it perfect on the first try. The goal is to improve measurably from Draft 1 to Draft 3. That's what universities look for: intellectual growth. [Pause] Questions?"*	LAYOUT: Large inspirational quote layout "Measurable Improvement" or "Intellectual Growth" as hero text Dark blue background, white/gold text Bookend with Slide 1 (professional, impactful close) Leave space for "Questions?" at bottom

FULL SPEAKER SCRIPT

Print this section separately and bring to the workshop. Timing cues in [brackets].

Opening Hook [00:00-05:00]

"Show of hands: Who here has written a science essay that your teacher said was 'too technical' or 'hard to follow'? [Pause for hands]Now, different question: Who has read a science article online and thought, 'This is dumbed down—they're treating me like I'm stupid'? [Pause]That's the translation problem. Chemistry has three audiences:1. Experts (your chemistry teacher, university professors)2. Educated non-specialists (A-Level students in OTHER subjects)3. General public (your parents, younger students)Most of you write for Audience 1 because that's who grades your exams. But CATS Insights publishes for Audience 2. That's the skill we're learning today: how to explain complex chemistry clearly without oversimplifying.The article I'm about to show you—on the White Cliffs—is written at exactly this level. Let's reverse-engineer how it works."

Teaching Block 1: The Framework [05:00-15:00]

"Here's the framework. [Point to table]Level 1 is what you write in exams. It assumes your reader already knows the jargon. 'Rhombohedral cleavage,' 'trigonal crystal system'—your chemistry teacher knows what these mean. You don't need to explain them. But nobody else does.Level 2 is our target. Technical but explained. Notice how it keeps the word 'trigonal' but immediately explains what it means: 'imagine the atoms arranged in a three-dimensional lattice.' It assumes curiosity, not expertise.Level 3 is oversimplified. 'Tiny sea creatures' is cute, but it's not chemistry. It treats the reader like they can't handle the real explanation.[Move to techniques]Four techniques for Level 2 writing. One: Define on first use. Two: Use analogies. Three: Connect to real world. Four: Explain the 'so what.'"

Exercise Launch [15:00-15:30]

"Now it's your turn. You have 10 minutes. On your handout, there's a Level 1 paragraph from the White Cliffs article. Your job: rewrite it at Level 2. Use the four techniques we just covered. Work with the people at your table. I'll walk around and help."

Circulating Prompts [15:30-25:00]

[Use these when coaching groups]• "Would your History A-Level friend understand this?"• "You've kept the fact—now explain WHY it matters."• "Good—you removed jargon. But did you replace it with clarity or vagueness?"• "Read it out loud. Does it sound natural?"• "What's the 'so what' here? Why should a non-chemist care?"

Debrief & Quality Standards [25:00-35:00]

"Okay, time's up. Let's hear some rewrites. [Call on 2-3 groups][After each group shares, apply the Three Tests publicly]'The Specialist Test: Would a chemistry teacher correct any facts here? [Discuss] The Peer Test: Could a History student follow this? [Discuss] The Citation Test: Are all claims backed by sources? [Usually no—that comes later]'Before anything goes on CATS Insights, it passes these three tests. If your draft fails any of these, the Lead Editor sends it back for revision. That's not punishment—it's mentorship. We're training you to write at university level."

Next Steps & Close [35:00-45:00]

"By next week's meeting:1. Choose a chemistry topic from the Topic Menu handout.2. Write a 300-word Level 2 explanation of one key concept from that topic.3. Submit via the Google Form—the link is in the Teams chat.Your draft will go through our editorial process:• Round 1: Peer review by another writer• Round 2: Feedback from Lead Editor (Grisha)• Round 3: My final approvalThe goal is

NOT to get it perfect on the first try. The goal is to improve measurably from Draft 1 to Draft 3. That's what universities look for: intellectual growth.Questions?"

Step 4

Good. You're confident on all four Kent topics — that's a strong position. You have Canva, which is the right tool for Instagram carousels. Here's your executable plan.

The 6-Post Content Calendar (8 Feb – 28 Feb)

The strategic logic: you need a mix of **brand-building** posts (so judges understand what CATS Insights is) and **chemistry content** posts (so judges see you promoting chemistry to your community). Two posts per week. Here's the schedule:

#	Post Date	Type	Topic	RSC Category Hit
1	**9 Feb (Sun)**	Carousel (5 slides)	"What is CATS Insights?"	Cat. 5
2	**13 Feb (Thu)**	Carousel (7 slides)	Chemistry of Kent: White Cliffs	Cat. 2 + 5
3	**17 Feb (Mon)**	Carousel (5 slides)	Meet the Team / Behind the Scenes	Cat. 4 + 5
4	**20 Feb (Thu)**	Carousel (7 slides)	Chemistry of Kent: Pfizer Sandwich	Cat. 2 + 5
5	**24 Feb (Mon)**	Single Image + Long Caption	Workshop Announcement / Recap	Cat. 3 + 5
6	**27 Feb (Thu)**	Carousel (7 slides)	Chemistry of Kent: Hop Chemistry	Cat. 2 + 5

Post 1: "What is CATS Insights?" — Start This Weekend

This is your foundation post. Every subsequent post makes more sense if this exists first. It tells new followers (and judges) what they're looking at.

Format: 5-slide Canva carousel, 1080×1080px, Harrow Blue background with white/gold text.

Slide 1 (Hook — must stop the scroll)

- **Visual:** Bold white text on Harrow Blue (#0C2340) background. CATS Insights logo if you have one.
- **Text on slide:**

  We're building something different.

  A student-led STEM publication with real academic standards.

Slide 2 (The Problem)

- **Visual:** Clean layout, one key stat or statement.
- **Text on slide:**

  The Problem

  A-Level students engage with science in exams — but rarely get the chance to research, write, and publish like real scientists.

  We're changing that.

Slide 3 (The Solution)

- **Visual:** 3 short bullet points with icons (Canva has good icon libraries).
- **Text on slide:**

  What is CATS Insights?

  📝 A student-led academic publication

  🔬 Covering Biology, Chemistry, Maths & more

  🎓 Written for educated non-specialists — not just experts

Slide 4 (The Standards)

- **Visual:** This is where you flex the rigour. Use a contrasting accent (Harrow Gold #89764B) for emphasis.
- **Text on slide:**

  Our Standards

  ✅ Oxford Referencing with DOI citations

  ✅ Turnitin plagiarism screening (<20%)

✅ AI detection via GPTZero

✅ Structured editorial review pipeline

This isn't a school newsletter.

Slide 5 (CTA — Call to Action)

- **Visual:** Clean, inviting. Include your school name.
- **Text on slide:**

  Based at The Worthgate School, Canterbury

  Follow us for chemistry content, research insights, and behind-the-scenes of student academic publishing.

  🔗 More coming soon.

Caption:

Most school science stops at the exam. We think it shouldn't.

CATS Insights is a student-led STEM publication at The Worthgate School, Canterbury. We train students to research, write, and publish academic articles — with real editorial standards including Oxford referencing, plagiarism detection, and structured peer review.

We're not a school newsletter. We're building something that lasts.

More chemistry content coming soon. 🔬

#STEMEducation #YoungChemist #ChemistryCommunication #StudentResearch #KentSchools #AcademicWriting #ScienceCommunication #CATSInsights

---

Post 2: "The Chemistry of the White Cliffs of Dover" — Publish 13 Feb

This is your first proper Kent chemistry piece. It must be **scientifically accurate** and **visually clean**. This is RSC evidence.

Format: 7-slide Canva carousel, 1080×1080px.

Slide 1 (Hook)

- **Visual:** A striking image of the White Cliffs (use a royalty-free image from Unsplash/Pexels). Bold overlay text.
- **Text:**

  The Chemistry Behind Kent's Most Famous Landmark

  The White Cliffs of Dover 🧪

Slide 2 (What Are They Made Of?)

- **Text:**

  What gives them that white colour?

  The cliffs are composed almost entirely of **calcium carbonate ($CaCO_3$)** — specifically a soft, fine-grained form called **chalk**.

  This chalk formed from the skeletal remains of **coccolithophores** — microscopic marine algae — that accumulated on the seabed over ~70 million years during the Late Cretaceous period.

Slide 3 (The Key Reaction)

- **Visual:** Display the equation prominently in a "formula card" style.
- **Text:**

  The Chemistry of Erosion

  Rainwater absorbs atmospheric CO_2 to form **carbonic acid**:

  $$\mathbf{CO_2(g) + H_2O(l) \rightarrow H_2CO_3(aq)}$$

  This weak acid reacts with calcium carbonate:

  $$\mathbf{CaCO_3(s) + H_2CO_3(aq) \rightarrow Ca(HCO_3)_2(aq)}$$

  The insoluble chalk dissolves into soluble **calcium bicarbonate** — and the cliff retreats.

Slide 4 (Rate of Erosion)

- **Text:**

How fast are the cliffs disappearing?

Average erosion rate: **0.5–1.0 cm per year**

But climate change is accelerating this. Rising CO_2 levels increase the acidity of rainfall, speeding up the dissolution reaction.

Some sections have lost **over 10 metres** in a single landslip event.

Slide 5 (The Thermal Decomposition Link)

- **Text:**

  A-Level Connection: Thermal Decomposition

  Heat calcium carbonate and you get a reaction every Chemistry student knows:

  $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

  This produces **quicklime (calcium oxide)** — used in cement, steelmaking, and water treatment.

  Kent's medieval lime kilns used chalk from these very cliffs.

Slide 6 (Environmental Significance)

- **Text:**

  Why This Matters

  The White Cliffs are a **carbon reservoir**. The $CaCO_3$ locked in the chalk represents millions of years of sequestered carbon.

  As erosion releases $Ca(HCO_3)_2$ into seawater, some CO_2 is ultimately returned to the atmosphere — creating a slow-motion feedback loop in the global carbon cycle.

Slide 7 (Branding slide)

- **Visual:** CATS Insights branding, Harrow Blue background.
- **Text:**

  The chemistry is beneath your feet.

  CATS Insights — real science, real writing, real Kent.

Follow for more 🔬

Caption:

You've seen the White Cliffs of Dover. But do you know the chemistry that built them — and the chemistry that's destroying them?

70 million years of microscopic marine organisms created Kent's most iconic landmark. Now, carbonic acid from atmospheric CO_2 is slowly dissolving it, one rainfall at a time.

Swipe through for the full breakdown — from coccolithophore skeletons to the global carbon cycle. ⬅️

Save this post if you're studying AQA/OCR Chemistry — the thermal decomposition of $CaCO_3$ is core content.

#ChemistryOfKent #WhiteCliffsOfDover #CalciumCarbonate #ALevelChemistry #ScienceCommunication #KentHistory #CATSInsights #RSCChemistry #GreenChemistry #CarbonCycle

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Posts 3–6: Brief Outlines

I'll flesh these out fully when you're ready for each one, but here's what you're building toward:

Post 3 (17 Feb) — "Meet the Team": Photo carousel of your leadership team (Alex, Grisha, Philip, Cindy, etc.) with their role titles and one sentence each about why they joined. Humanises the brand and shows the mentoring structure for Cat. 4.

Post 4 (20 Feb) — "Pfizer Sandwich": Same 7-slide chemistry carousel format. Cover Pfizer's 1954 establishment in Sandwich, the Viagra/sildenafil discovery story, the site's closure and rebirth as Discovery Park, and the medicinal chemistry of drug development. This is your strongest "industrialist judge" content.

Post 5 (24 Feb) — Workshop Post: If workshop has happened, post a recap with photos. If it hasn't yet, post an announcement teaser. Either way, this is Cat. 3 evidence on your feed.

Post 6 (27 Feb) — "Kent Hop Chemistry": Cover alpha-acid (humulone) isomerisation during brewing, beta-acids and oxidation, Kent's "Garden of England" heritage, and the organic chemistry of flavour compounds. This is the most unusual topic — judges will remember it.

Metrics Tracking

Create a Google Sheet with these columns and fill it after each post hits 48 hours:

Date	Post #	Topic	Type	Reach	Impressions	Saves	Shares	Comments	Follower Count

Screenshot the Instagram Insights page for each post and save to `RSC_Evidence/Digital_Outreach/Instagram_Screenshots/`.

Your next action right now: Open Canva, create a new 1080×1080px design, set the background to #0C2340 (Harrow Blue), and start building Post 1 slides. You can have it designed and posted by tomorrow evening. Want me to build the full slide-by-slide content for the Pfizer Sandwich or Kent Hops post next?

To Do List

Schedule and get venue for the workshop for after half term break
Get le to make the presentation slides over the break
Prepare media and other parts (memorise your bits) for the workshop
Find a date for the presentation and perform it

Jason Ver Step 6-12

RSC KENT YOUNG CHEMIST AWARD

STEPS 6–17: THE COMPLETE EXECUTION MANUAL

From Evidence Archiving to Final Submission

Prepared for: Jason | President, CATS Insights
School: The Worthgate School, Canterbury, Kent
Document Date: 4 March 2026
Submission Deadline: 31 March 2026
Days Remaining: 27 Days
Phase Status: Phase 2 Completion → Phase 3 (PACKAGE) → Phase 4 (SUBMIT)
Document Version: 3.0 | Steps 6–17 Detailed Execution

"You have 27 days. Every deliverable in this document has a hard deadline. Miss one, and the chain breaks."

MASTER TIMELINE: 4 March – 28 March 2026

Today is **Wednesday 4 March 2026**. Your teacher submits the Google Form by **28 March** (3 days before the 31 March hard deadline). You have **24 working days** to complete Steps 6–17. Below is the compressed, realistic timeline recalibrated from today.

Step	Deadline	Deliverable	RSC Category
6	4–7 Mar	Screenshot & archive all existing posts + create Metrics Log	Cat. 5 (Evidence)
7	5 March	Workshop delivered + all evidence archived	Cat. 3, 4
8	4–9 Mar	Mentoring Impact Tracker + before/after samples + testimonials	Cat. 4
9	4–7 Mar	Lead Writer feedback system documented + evidence trail saved	Cat. 4
10	5–10 Mar	Chemistry infographic designed, posted, shared with dept	Cat. 1
11	10–14 Mar	Impact Report PDF (2–3 pages)	All
12	14–17 Mar	Evidence folder compiled on Google Drive	All
14	15–20 Mar	Project video filmed, edited, uploaded (YouTube Unlisted)	Cat. 5
16	24–27 Mar	Final QA on all links, permissions, PDFs	—

CRITICAL REALITY CHECK

Steps 6–9 should already be partially complete from Phase 2. If they are NOT, you are behind schedule. Steps 6–9 run IN PARALLEL from 4–9 March. Steps 10–14 overlap from 5–20 March.

STEP 6: Screenshot & Archive Everything

Deadline: 4–7 March 2026 | Owner: You + Duc (Secretary)

RSC Category: Evidence backbone for Category 5 (Digital Outreach)

This is not a creative step. It is an administrative discipline step. Without systematic evidence capture, every Instagram post, every workshop photo, and every mentoring document is invisible to the judges. The judges will never see your Instagram account directly. They will see **screenshots in a Google Drive folder**. That folder must be immaculate.

6.1 Instagram Evidence Capture

6.1.1 Post-Level Screenshots

For **every Instagram post** published during the content sprint (Posts 1–6 minimum), capture the following screenshots **48–72 hours after publication** (this gives engagement time to stabilise):

- ☐ The post itself (full carousel or single image as it appears in the feed)
- ☐ Instagram Insights: Overview page showing Reach, Impressions, Profile Visits
- ☐ Instagram Insights: Content Interactions showing Likes, Comments, Saves, Shares
- ☐ If carousel: Swipe-through rate (percentage who viewed all slides)
- ☐ Any notable comments from educators, scientists, or external accounts

6.1.2 Account-Level Screenshots

Capture these **once per week** during the sprint and **once on the final day before packaging** (around 12 March):

- ☐ Total follower count (screenshot the profile page)
- ☐ Instagram Insights: Accounts Reached (7-day and 30-day view)
- ☐ Instagram Insights: Accounts Engaged (7-day and 30-day view)
- ☐ Top-performing content by reach
- ☐ Audience demographics if available (age, location)

6.1.3 The Metrics Log Spreadsheet

Create a Google Sheet titled **"RSC_Metrics_Log"** with these exact columns. This becomes Exhibit A in your evidence folder.

Date	Post #	Topic	Type	Reach	Impress.	Saves	Shares
e.g. 9 Feb	Post 1	What is CATS?	Carousel	142	389	12	5
13 Feb	Post 2	White Cliffs	Carousel	...	...	...	...

Add additional columns for: Follower Count at time of posting, Comments count, Engagement Rate (calculated as (Likes + Comments + Saves + Shares) / Reach × 100).

6.2 Blog/Article Evidence Capture

For each "Chemistry of Kent" long-form article (White Cliffs, Pfizer Sandwich, etc.):

- ☐ Save the final article as a PDF with full references visible
- ☐ Screenshot the article's publication location (Instagram caption, blog page, Google Doc)
- ☐ If published online: capture a Wayback Machine archive or web.archive.org snapshot as permanent proof
- ☐ Save the original Google Doc link with edit history intact (proves human authorship)

6.3 Google Drive Folder Structure

Create this folder structure **today** if it does not already exist. Do not wait until Step 12.

```
RSC_Evidence/
├── Digital_Outreach/
│   ├── Instagram_Screenshots/
│   │   ├── Post1_WhatIsCATS/
│   │   ├── Post2_WhiteCliffs/
│   │   ├── Post3_MeetTeam/
│   │   ├── Post4_PfizerSandwich/
│   │   ├── Post5_Workshop/
│   │   └── Post6_HopChemistry/
│   ├── Account_Level_Screenshots/
│   ├── Metrics_Log.xlsx
│   └── Kent_Chemistry_Articles/
├── Workshop_Evidence/
├── Mentoring_Evidence/
├── Teaching_Resource/
└── Foundational_Documents/
```

STEP 6 COMPLETION GATE

☐ Metrics Log spreadsheet created with all posts logged
☐ Post-level screenshots for every published post saved in correct subfolders
☐ Account-level screenshots captured (at least 2 snapshots: mid-sprint + final)
☐ All Kent Chemistry articles saved as PDFs with references
☐ Google Drive folder structure created and populated

IF ANY OF THESE ARE MISSING, STOP. Do not proceed to Step 11.

STEP 7: Workshop Evidence Consolidation

Deadline: By 7 March 2026 | Owner: You + Cindy + Grisha
RSC Categories: 3 (STEM School Event) + 4 (Mentoring)

STATUS CHECK
The workshop ("Writing Chemistry for Non-Chemists") was scheduled for the week of 17–21 February.
If it has been delivered: this step is about consolidating and packaging the evidence.
If it has NOT been delivered: you must deliver it THIS WEEK (by 7 March) or you lose Category 3 entirely.

7.1 If Workshop Is Complete: Evidence Packaging

7.1.1 Attendance Documentation

- ☐ Scan or photograph the attendance sheet → save as PDF
- ☐ Create a summary line: "[X] students from Years [Y–Z] attended the 45-minute workshop on [DATE]"
- ☐ If you used a Google Form for attendance, export the responses as a CSV/PDF

7.1.2 Media Consolidation

- ☐ Collect ALL photos from Duc/Aung/Cindy — get them from every phone that recorded anything
- ☐ Select the 5–10 best photos: wide shot, teaching moment, group work, student sharing, group photo
- ☐ Rename photos systematically: Workshop_01_RoomSetup.jpg, Workshop_02_Teaching.jpg, etc.
- ☐ Compile video clips into a single 2–3 minute highlight reel OR save as labelled individual clips
- ☐ Save all media to RSC_Evidence/Workshop_Evidence/Photos/ and .../Video_Clips/

7.1.3 Workshop Materials Archive

- ☐ Save the 3-Level Framework Handout as PDF
- ☐ Save the "Fix This Paragraph" Exercise Sheet as PDF
- ☐ Save the Topic Menu Handout as PDF
- ☐ Save the presentation slides as PDF (export from Canva/PowerPoint)
- ☐ Save these to RSC_Evidence/Workshop_Evidence/Workshop_Materials/

7.1.4 Homework Collection Status

- ☐ Export all Google Form submissions from the homework task
- ☐ Count total submissions vs. total attendees → calculate compliance rate
- ☐ If compliance is below 70%: send a FINAL reminder via Teams today with a 48-hour extension
- ☐ Tag the top 3 submissions (best Level 2 quality) for RSC before/after evidence

7.1.5 Workshop Impact Report

Write a 1-page Workshop Impact Report (this feeds into the main Impact Report at Step 11). Structure:

Title: CATS Insights Workshop: "Writing Chemistry for Non-Chemists"

Date: [DATE] | **Duration:** 45 minutes | **Facilitator:** Jason Ye Yint

Attendees: [X] CATS Insights members (Years [Y–Z])

Learning Objectives: (1) Distinguish between 3 audience levels. (2) Apply PEEL structure. (3) Self-assess using Three-Test Framework.

Quantitative Outcomes: [X]% of attendees submitted 300-word drafts by deadline. Average jargon reduction across submissions: [Y]%. [Z] students improved from Level 1 to Level 2 writing.

Qualitative Feedback: Include 2–3 direct student quotes about what they learned.

Measurable Improvement Sample: Student A Draft 1: 12 undefined technical terms, avg sentence length 38 words. Student A Draft 2: 3 terms (all defined), avg sentence length 19 words. Clarity improvement: +65%.

7.2 If Workshop Has NOT Been Delivered

You are in an emergency. Category 3 is worth a significant portion of your multi-category strategy.

Emergency Option A: Deliver it by 5 March

Non-Negotiable: The workshop MUST happen before 10 March

STEP 8: Document the Mentoring Pipeline

Deadline: 4–9 March 2026 | Owner: You + Grisha (Lead Editor)

RSC Category: 4 (Mentoring)

This step transforms informal mentoring activity into **quantifiable, judge-ready evidence**. The judges do not care that you "helped students write better." They care that you can **prove it with data**.

8.1 Create the Mentoring Impact Tracker

Create a Google Sheet titled **"Mentoring_Impact_Tracker"** with the following columns:

Student (Anon.)	Topic	Draft 1 Date	Draft 1 Issues	Feedback Given	Draft 2 Date	Improvement	Status
Student A	Hop Chem.	19 Feb	8 jargon terms, no PEEL, L1 writing	3-Level rewrite, PEEL structure	26 Feb	2 jargon, PEEL used, L2	Editorial-ready
Student B	Maillard Rxn	20 Feb	Level 3, oversimplified, no citations	Add citations, increase technical depth	28 Feb	L2 achieved, 3 citations added	Needs revision

8.2 Before/After Writing Samples

For a **minimum of 3 students**, prepare a side-by-side comparison document showing their Draft 1 versus their most recent draft. This is your most powerful evidence.

8.2.1 How to Prepare Each Comparison

- ☐ Select students who showed the MOST measurable improvement (not necessarily the best writers)
- ☐ Anonymise all names: use "Student A", "Student B", "Student C"
- ☐ For each student, create a 1-page PDF with this structure:

Header: "Mentoring Case Study: Student [X]"

Left column: Draft 1 excerpt (200–300 words) with Level classification and issue annotations

Right column: Revised draft excerpt with improvements highlighted

Bottom box: Quantitative metrics table:

Metric	Draft 1	Draft 2 (Post-Mentoring)
Unexplained jargon terms	8	2
Average sentence length (words)	38	19

PEEL structure adherence	0%	80%
Writing Level (1/2/3)	Level 1	Level 2
Citations included	0	3

8.3 Collect Student Testimonials

You need **3–5 quotes** from students who participated in the mentoring process. These appear in the Impact Report and nomination text.

8.3.1 How to Collect

☐ Send a Microsoft Form or Google Form to all workshop attendees and writers with these 3 questions:

1. "What did you learn from the CATS Insights editorial process?"
2. "How has the feedback you received improved your writing?"
3. "Would you recommend CATS Insights to other students? Why?"

☐ Set a **48-hour deadline** (responses needed by 6 March)

☐ Follow up personally via Teams DM with any non-responders after 24 hours

☐ Select the 3–5 strongest quotes and save them in a "Student_Testimonials.pdf" document

8.3.2 Example Strong Testimonials (Use As Quality Benchmark)

"Before CATS Insights, I didn't know how to structure a scientific argument. Now I can write a PEEL paragraph from memory."

"The editorial feedback taught me to explain chemistry for non-specialists — I use that skill in every subject now."

"I thought simplifying science meant dumbing it down. The 3-Level Framework showed me it's about clarity, not compromise."

STEP 8 COMPLETION GATE

☐ Mentoring Impact Tracker spreadsheet complete with at least 3 student rows
☐ Before/After writing comparison PDFs created for 3 students (anonymised)
☐ Quantitative metrics calculated for each comparison
☐ 3–5 student testimonial quotes collected and saved as PDF
☐ All saved to RSC_Evidence/Mentoring_Evidence/

STEP 9: Formalise the Lead Writer Feedback System

Deadline: 4–7 March 2026 | Owner: You + Grisha

RSC Category: 4 (Mentoring) — Systemic Evidence

Step 8 proves mentoring happened with individual students. Step 9 proves you built a **system** for mentoring that can continue without you. This is the institutional legacy angle that separates a Gold entry from a Silver.

9.1 Document the Workflow

Create a 1-page document titled **"CATS Insights Editorial Feedback Workflow"** that shows the complete pipeline:

Stage	Action	Owner
Stage 1	Writer submits Draft 1 via Google Form/Google Drive	Writer
Stage 2	Lead Editor (Grisha) reviews using 3-Level Framework + PEEL checklist	Grisha
Stage 3	Written feedback returned in Teams thread using Feedback Summary Template	Grisha
Stage 4	Writer revises and submits Draft 2	Writer
Stage 5	Final review: marked as "editorial-ready" or "needs further revision"	You / Grisha
Stage 6	Editorial-ready articles enter QA pipeline (Turnitin, GPTZero, citation check)	QA Team

9.2 Save the Evidence Trail

The feedback system only counts as evidence if the judges can **see it in action**. You need:

- ☐ At least 3 examples of written feedback given to writers (screenshots of Teams threads or Google Doc comments)
- ☐ At least 2 examples showing the revision cycle (Draft 1 → Feedback → Draft 2)
- ☐ The Feedback Summary Template filled in for at least 2 articles
- ☐ Screenshot of Google Drive showing the organised article folders with Draft 1, Feedback, Draft 2 visible

9.3 Create the Process Diagram

Create a simple visual flowchart (Canva or Google Slides) showing the 6-stage pipeline. This goes into the Impact Report and is powerful visual evidence. Use the CATS Insights brand colours (Harrow Blue #0C2340, Gold #89764B). Label each stage clearly. Include the role responsible at each stage.

Save as: RSC_Evidence/Mentoring_Evidence/Editorial_Workflow_Diagram.pdf

STEP 10: Produce the Chemistry Teaching Infographic

Deadline: 5–10 March 2026 | Owner: You (content) + Nana (design)

RSC Category: 1 (Non-Core Practical — Teaching Resource)

This is your **only evidence for Category 1**. Without this infographic, you have a gap. The specification says: "Describe the experiment and create a resource for teaching, preferably including a video or infographic." You are delivering an infographic.

10.1 Topic Selection

Choose **ONE** of the following. My recommendation: **The Chemistry of Calcium Carbonate** because it ties directly to your White Cliffs content (creating a narrative thread for judges) and covers A-Level core content (thermal decomposition, acid-base reactions).

Option	Content	Judge Appeal
$CaCO_3$ Chemistry	Thermal decomposition, acid reactions, carbon cycle, White Cliffs tie-in	Academic + Educator + Kent link
Green Chemistry: 12 Principles	All 12 principles with industrial examples, classroom display format	Industrialist + Educator
Molecular Chirality	R/S configuration, thalidomide case study, pharmaceutical relevance	Academic + Industrialist

10.2 Infographic Specifications

10.2.1 Format & Dimensions

Size: A3 portrait (297mm × 420mm) for classroom display; also export a 1080×1350px version for Instagram

Tool: Canva (use the "Infographic" template category as starting point)

File formats: Export as PDF (print-quality, 300 DPI) + PNG (Instagram)

10.2.2 Content Structure (Calcium Carbonate)

The infographic must contain ALL of the following sections:

Section 1: Title & Hook

"The Chemistry of Calcium Carbonate: From White Cliffs to Your Classroom"

CATS Insights logo in top corner. Subtitle: "A Teaching Resource for A-Level Chemistry"

Section 2: What Is $CaCO_3$?

Chemical formula, molecular structure diagram, crystal system (trigonal/calcite), physical properties (white, soft, insoluble in water). Ball-and-stick or space-filling model image.

Section 3: Key Reactions (3 reaction equations)

Reaction 1 — Thermal Decomposition: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ [with temperature: ~825°C]

Reaction 2 — Acid Reaction: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

Reaction 3 — Weathering: $CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$

Each equation must be clearly formatted with state symbols. Use a "reaction card" visual for each.

Section 4: Real-World Applications

Cement/construction (CaO from thermal decomposition), agriculture (neutralising acidic soil), antacids (neutralisation), carbon cycle (geological sequestration). Use icons for each application.

Section 5: The Kent Connection

White Cliffs of Dover: 100 million years of coccolithophore accumulation. Medieval lime kilns: industrial quicklime production. Include a small map of Kent with Dover marked.

Section 6: Footer/Branding

"Produced by CATS Insights | The Worthgate School, Canterbury" — CATS Insights logo, @catsinsights handle, date.

10.2.3 Design Requirements

Colours: Harrow Blue (#0C2340) headers, Harrow Gold (#89764B) accent lines, white background, dark navy (#1A1A1A) body text

Typography: Montserrat Bold for headings (18–24pt), Open Sans for body text (10–14pt)

Equations: Rendered clearly using Canva text or LaTeX PNG imports (minimum 300 DPI). State symbols must be visible.

Visual hierarchy: Title → Structure → Reactions → Applications → Kent Connection. Reader should scan top-to-bottom in 60 seconds.

Accessibility: All text readable from 1.5 metres (classroom wall distance). Minimum font size 10pt for body text.

10.3 Evidence Capture (Critical)

The infographic itself is only half the evidence. You also need proof that it was **used as a teaching resource**:

- ☐ Print 2 copies on A3 paper (use school printer or local print shop)
- ☐ Give one copy to your Chemistry teacher with a brief note: "I created this as a teaching resource for CATS Insights. You are welcome to display it in the classroom or use it in lessons."
- ☐ Photograph the teacher receiving it or the infographic displayed in the classroom
- ☐ Ask the teacher to send you a brief acknowledgement message (email or Teams) — even one sentence is sufficient: "Thank you for the infographic, I will use it with my students."
- ☐ Screenshot the acknowledgement message → save as Teacher_Acknowledgement.png
- ☐ Post the Instagram version on the CATS Insights account with caption explaining it as a teaching resource
- ☐ Screenshot the Instagram post + analytics after 48 hours

10.4 Delegation Timeline

Date	Task	Detail	Owner
5 Mar (Wed)	Content brief	Write all text, equations, section content	You
6 Mar (Thu)	Design start	Nana begins Canva layout using brief	Nana
8 Mar (Sat)	First draft	Nana sends design for your review	Nana → You
9 Mar (Sun)	Accuracy review	You verify all equations, facts, formatting	You
10 Mar (Mon)	Final export	Export PDF + PNG, print A3 copies	Nana + You
10–11 Mar	Deliver + post	Give to teacher + post on Instagram	You

STEP 10 COMPLETION GATE

☐ Infographic designed with all 6 sections, branded correctly, equations accurate
☐ Exported as A3 PDF (print) and 1080×1350 PNG (Instagram)
☐ Printed on A3 and delivered to Chemistry teacher
☐ Teacher acknowledgement message received and screenshotted
☐ Posted on Instagram with teaching resource caption
☐ Post analytics screenshotted after 48 hours
☐ All saved to RSC_Evidence/Teaching_Resource/

STEP 11: Write the Impact Report

Deadline: 10–14 March 2026 | Owner: You (sole author)

RSC Category: ALL — This is the centrepiece of your submission

WHY THIS MATTERS

The Impact Report is the single document that tells judges: "Here is everything I did, here is the evidence, and here is why it matters."

It is NOT a summary of CATS Insights. It is a targeted, RSC-aligned presentation of impact.

Every claim must be backed by a number, a screenshot, or a student quote. No vague assertions.

11.1 Document Specifications

Title: "CATS Insights: Impact Report for RSC Kent Young Chemist Award 2026"

Length: 2–3 pages (A4). Do not exceed 3 pages. Judges have limited time.

Format: PDF. Design in Canva or Google Docs with CATS branding.

Tone: Professional, evidence-driven, third person ("The Founder" not "I"). Include RSC keywords throughout.

11.2 Page-by-Page Structure

PAGE 1: Executive Summary + Category Mapping

Section A: Executive Summary (200–250 words)

Who: Jason Ye Yint, Year 12, Chemistry A-Level, The Worthgate School, Canterbury.

What: Founded CATS Insights — a student-led STEM publication with professional editorial standards.

Problem: A-Level students had no platform to engage with real academic research beyond the syllabus.

Solution: Built a publication with Oxford referencing, Turnitin screening (<20%), GPTZero AI detection, structured mentoring pipeline, and a 27-member team across strategic, operational, and execution tiers.

Scale: [X] active contributors, [Y] articles in development, [Z] Instagram posts reaching [N] accounts.

Section B: RSC Category Mapping Table

This table must appear on Page 1. It tells judges exactly where to find evidence for each category:

#	RSC Category	CATS Insights Activity	Key Evidence
1	Teaching Resource	$CaCO_3$ Infographic for classroom use	PDF, teacher acknowledgement, Instagram post
2	History of Chemistry	"Chemistry of Kent" series (White Cliffs, Pfizer, Hops)	Published articles, Instagram carousels
3	STEM School Event	"Writing Chemistry for Non-Chemists" Workshop	Attendance, photos, video, materials

4	Mentoring	Lead Writer system + editorial feedback cycles	Before/after samples, testimonials, tracker
5	Digital Outreach	CATS Insights Instagram + Chemistry of Kent blog	Metrics log, screenshots, engagement data

PAGE 2: Evidence of Impact

Section C: Workshop Outcomes

Attendance: [X] students, Years [Y–Z], [DATE], 45 minutes.

Pre-workshop: [X]% of participants could not identify the difference between Level 1 and Level 2 writing.

Post-workshop: [Y]% successfully rewrote a technical paragraph to Level 2 standard.

Follow-up: [Z]% of attendees submitted 300-word drafts within 7 days.

Include 1–2 student quotes from Section 8.3.

Section D: Mentoring Metrics

Number of writers mentored: [X] active contributors in the editorial pipeline.

Average improvement: Jargon reduction of [Y]%, sentence length reduction of [Z]%.

Sample case: "Student A improved from 8 unexplained technical terms (Draft 1) to 2 terms, all defined in context (Draft 2), with PEEL adherence increasing from 0% to 80%."

Include reference to Before/After Samples PDF in evidence folder.

Section E: Digital Reach

Total posts published: [X] chemistry-focused posts over [Y] weeks.

Total reach: [N] unique accounts reached.

Engagement metrics: [X] saves, [Y] shares (these indicate high-value content).

Follower growth: From [A] to [B] (Δ = [C] followers, [D]% growth).

Content themes: Chemistry of Kent series (White Cliffs, Pfizer Sandwich, Hop Chemistry) connecting local community to chemistry.

PAGE 3: Institutional Legacy

Section F: Governance & Sustainability

Organisational structure: 27-member team across 3 tiers (Strategic, Operational, Execution).

Foundational Document Library: 48 documents across 7 suites including Governance, Editorial Standards, QA Protocols, and Production Workflows.

Succession plan: Every leadership role has a designated deputy. The "Succession Test" ensures a new team could publish Edition 1 within 6 months using documented systems alone.

Quality assurance: 4-stage audit pipeline — Technical Edit, Line Edit, QA Gate (colour-coded: GREEN/AMBER/RED/BLACK), Final Approval.

Section G: Closing Statement

CATS Insights is not a personal project; it is an institutional platform designed to promote chemistry to a wider audience and democratise access to academic publishing. The systems, standards, and training programmes built ensure this work outlasts any single student. This is leadership through education.

11.3 RSC Keywords Checklist

Before finalising, CTRL+F the Impact Report for each keyword. Every one must appear at least once:

- ☐ Academic excellence — appears in Impact Report
- ☐ Promotion of chemistry — appears in Impact Report
- ☐ Creativity and latest technology — appears in Impact Report
- ☐ Community engagement — appears in Impact Report
- ☐ Mentoring impact — appears in Impact Report
- ☐ Diversity and accessibility — appears in Impact Report
- ☐ Sustainability — appears in Impact Report
- ☐ Institutional legacy — appears in Impact Report
- ☐ Role models — appears in Impact Report

STEP 12: Compile the Evidence Folder

Deadline: 14–17 March 2026 | Owner: You (with Cindy assisting)

RSC Category: ALL — This is the submission backbone

This step converts your scattered Google Drive files into a **single, judge-ready evidence portfolio**. Think of it as a legal exhibit file. Every document labelled, every folder logical, every link working.

12.1 Final Folder Structure

Create a **new Google Drive folder** titled exactly: **RSC_Young_Chemist_Submission**

This is the folder you will share with your teacher. It must be self-contained and navigable by someone who has never seen your work before.

```
RSC_Young_Chemist_Submission/
├── 00_Impact_Report.pdf
├── 01_Digital_Outreach/
│   ├── Instagram_Screenshots/ (subfolder per post)
│   ├── Account_Analytics/ (weekly + final snapshots)
│   ├── Metrics_Log.xlsx
│   └── Kent_Chemistry_Articles/ (PDFs with references)
├── 02_Workshop_Evidence/
│   ├── Attendance_Sheet.pdf
│   ├── Workshop_Materials/ (handouts, slides as PDFs)
│   ├── Photos/ (5–10 best, renamed systematically)
│   ├── Video_Clips/ (highlight reel or labelled clips)
│   └── Workshop_Impact_Report.pdf
├── 03_Mentoring_Evidence/
│   ├── Mentoring_Impact_Tracker.xlsx
│   ├── Before_After_Samples.pdf (3 anonymised case studies)
│   ├── Student_Testimonials.pdf
│   └── Editorial_Workflow_Diagram.pdf
├── 04_Teaching_Resource/
│   ├── CaCO3_Infographic.pdf (A3 print version)
│   ├── CaCO3_Infographic_Instagram.png
│   └── Teacher_Acknowledgement.png
└── 05_Foundational_Documents/
    ├── CATS_Insights_Proposal.pdf
    ├── Operations_Manual_Summary.pdf (1–2 page extract)
    ├── QA_Protocol_Summary.pdf (1-page extract)
    └── Organisational_Chart.pdf
```

12.2 Folder Compilation Checklist

Work through this systematically. Do not skip steps.

12.2.1 Digital Outreach (Folder 01)

- ☐ Copy all post screenshots from RSC_Evidence into 01_Digital_Outreach/Instagram_Screenshots/
- ☐ Copy account-level analytics screenshots into Account_Analytics/
- ☐ Copy the completed Metrics_Log.xlsx
- ☐ Copy all Kent Chemistry article PDFs (White Cliffs, Pfizer Sandwich, etc.)
- ☐ Verify: every file opens correctly, no broken links, no empty folders

12.2.2 Workshop Evidence (Folder 02)

- ☐ Copy attendance sheet PDF
- ☐ Copy all workshop handout PDFs (3-Level Framework, Fix This Paragraph, Topic Menu)
- ☐ Copy presentation slides as PDF
- ☐ Select and copy 5–10 best photos (renamed: Workshop_01, Workshop_02, etc.)
- ☐ Copy video clips or highlight reel
- ☐ Copy Workshop Impact Report

12.2.3 Mentoring Evidence (Folder 03)

- ☐ Copy Mentoring_Impact_Tracker.xlsx
- ☐ Copy Before_After_Samples.pdf (all 3 case studies in one document)
- ☐ Copy Student_Testimonials.pdf
- ☐ Copy Editorial_Workflow_Diagram.pdf
- ☐ Copy 2–3 screenshots of actual feedback given in Teams/Google Docs

12.2.4 Teaching Resource (Folder 04)

- ☐ Copy infographic PDF (A3 version)
- ☐ Copy infographic PNG (Instagram version)
- ☐ Copy Teacher_Acknowledgement.png

12.2.5 Foundational Documents (Folder 05)

- ☐ Copy or create CATS_Insights_Proposal.pdf (the original founding document)
- ☐ Create Operations_Manual_Summary.pdf: a 1–2 page extract highlighting governance structure, succession plan, and key protocols
- ☐ Create QA_Protocol_Summary.pdf: a 1-page extract showing the Turnitin, GPTZero, citation verification, and colour-coded scoring system
- ☐ Create Organisational_Chart.pdf: a visual showing the 3-tier structure with role names

12.3 Sharing Permissions

Before sharing the folder link with your teacher:

- ☐ Set the top-level folder to "Anyone with the link can view"
- ☐ Test EVERY subfolder and file by opening in an incognito browser window
- ☐ Verify all PDFs render correctly on mobile (judges may view on phones)
- ☐ Verify all video clips play without downloading
- ☐ Verify the Metrics Log opens correctly in Google Sheets

STEP 13: Draft the Teacher's Nomination Text

Deadline: 14–18 March 2026 | Owner: You (sole author)

This is the text your teacher submits in the Google Form

STRATEGIC PRINCIPLE

Your teacher will submit the nomination via Google Form. If you leave the writing to them, you lose control of the narrative.

Draft the full text yourself. Present it to the teacher for approval/edits. This is professional, not presumptuous.

The teacher's voice must come through — write it AS IF the teacher is speaking about you in the third person.

13.1 Nomination Text Specifications

Length: 400–500 words (the Google Form likely has a text limit — check it)

Voice: Written as the teacher speaking about the student. "I am nominating Jason Ye Yint..."

Tone: Professional, evidence-based, specific. No vague praise. Every claim backed by a number.

13.2 Mandatory Content Sections

Opening (50–70 words)

Introduce the student: name, year group, qualification, school. State what CATS Insights is in one sentence. State the nomination is for the Level 3 Young Chemist Award.

Categories Addressed (100–150 words)

Explicitly name each RSC category the entry covers. One sentence per category with one concrete example:

Category 1: "Jason created a professionally designed calcium carbonate infographic now used as a teaching resource in our Chemistry department."

Category 3: "He delivered a 45-minute workshop teaching [X] students to write chemistry for non-specialist audiences."

Category 4: "He mentored [X] student writers through a structured editorial feedback system, producing measurable improvements in writing quality."

Category 5: "He built and manages the CATS Insights Instagram account, publishing [X] chemistry-focused posts reaching [Y] accounts."

Impact Evidence (100–150 words)

Specific numbers only. No adjectives without data. Examples:

"Mentored [X] students, with an average jargon reduction of [Y]% between drafts."

"Published [X] posts reaching [Y] unique accounts, with [Z] saves indicating high-value content."

"Workshop attended by [N] students, with [M]% submitting follow-up work within 7 days."

"Organisation comprises 27 student members across 3 operational tiers."

Institutional Legacy (50–80 words)

"What sets Jason's work apart is its sustainability. He has created a Foundational Document Library of 48 documents including governance protocols, editorial handbooks, QA procedures, and a succession plan ensuring CATS Insights continues to operate after his departure. This demonstrates leadership through education, not personal ambition."

Closing (30–50 words)

"I wholeheartedly recommend Jason for the Gold certificate. He has demonstrated academic excellence, community engagement, mentoring impact, and institutional leadership that is exceptional for a student of his age."

13.3 RSC Keywords — Must Appear

Cross-reference these against the nomination text before finalising. Each must appear at least once: **academic excellence, promotion of chemistry, creativity, latest technology, community engagement, mentoring impact, accessibility, diversity, sustainability, role models.**

13.4 Teacher Delivery Process

- ☐ Draft the full 400–500 word text by 18 March
- ☐ Format it cleanly in a Google Doc with a note: "Draft nomination text for your review. Please amend as you see fit."
- ☐ Share the Google Doc with your teacher via email/Teams by 18 March
- ☐ Request feedback by 22 March (before the teacher review meeting)
- ☐ Be prepared for the teacher to make edits — this is their nomination, not yours

STEP 14: Film the Project Video

Deadline: 15–20 March 2026 | Owner: You (presenter) + Duc/Aung (camera)

RSC Category: 5 (Digital Outreach) + general submission asset

A 2–3 minute video linked in the nomination gives judges a visceral sense of your project that no PDF can replicate. Authenticity beats production value. A well-structured phone video is better than a polished but empty one.

14.1 Video Structure (2:30 target)

Timestamp	Content	Visuals
0:00–0:20	HOOK: "I'm Jason, and I founded CATS Insights to solve a problem: students at my school had no way to engage with real scientific research beyond the syllabus."	Direct to camera. Clean background (classroom or library). Good lighting.
0:20–1:00	WHAT IT IS: Explain CATS Insights. Mention team size (27 members), editorial standards (Turnitin, Oxford referencing), the 3-Level Writing Framework.	Cut to: Instagram feed on screen, team photo, workshop photo.
1:00–1:40	IMPACT: Workshop delivery, mentoring examples, specific numbers. "We trained [X] students in science communication. Our writers showed an average [Y]% improvement in clarity."	Cut to: Workshop footage, student work on screen, testimonial quote cards.
1:40–2:00	KENT CONNECTION: "We write about the chemistry happening in Kent — from the calcium carbonate of the White Cliffs to Pfizer's pharmaceutical legacy in Sandwich."	Cut to: White Cliffs Instagram carousel, Pfizer content, infographic.
2:00–2:30	LEGACY: "I built this to last. 48 foundational documents, a succession plan, and trained deputies in every role. The next generation of student editors can continue this work."	Cut to: Drive folder structure on screen, organisational chart. End on direct to camera.

14.2 Production Specifications

Equipment: Smartphone camera (1080p minimum). Tripod or stable surface. Ring light or natural window light.

Audio: Quiet room. No background music during speech. Consider using a lapel mic if available.

Editing: Use CapCut (free), iMovie, or similar. Simple cuts between segments. No flashy transitions. Add text overlays for key statistics (e.g., "27 team members", "[X]% clarity improvement").

Branding: Add CATS Insights logo as a corner watermark. Use Harrow Blue (#0C2340) for any text overlays.

Length: Strictly 2:00–2:30. Under 2 minutes feels thin. Over 3 minutes and judges stop watching.

14.3 B-Roll Footage Needed

Film these supplementary clips before or during the video shoot. Each should be 5–10 seconds:

- ☐ Scrolling through the CATS Insights Instagram feed on a phone
- ☐ The team working together (even a staged photo of group looking at laptops)
- ☐ The infographic printed and displayed
- ☐ The Google Drive evidence folder open on a laptop screen
- ☐ A close-up of a marked-up student draft with editorial comments visible
- ☐ The organisational chart on screen

14.4 Upload & Access

- ☐ Upload to YouTube as UNLISTED (not Private — Private requires sign-in to view)
- ☐ Title: "CATS Insights | RSC Young Chemist Award 2026 — Jason Ye Yint"
- ☐ Description: Brief 2-sentence summary + "The Worthgate School, Canterbury, Kent"
- ☐ Test the link in an incognito browser to confirm it plays without sign-in
- ☐ Save the YouTube link in a text file within the evidence folder

STEPS 15–17: The Submission Sequence

Phase 4: SUBMIT | 22–28 March 2026

THE FINAL SPRINT

Everything comes together here. Steps 15–17 are sequential and time-critical.
Your teacher submits the Google Form. You are the project manager ensuring every asset is ready.
Target submission: 26–28 March. NOT March 31. You need a buffer.

STEP 15: Teacher Review Meeting (22–24 March)

Owner: You | Duration: 15–20 minutes

This meeting has one objective: walk your teacher through the complete submission package so they can submit with confidence.

15.1 Meeting Preparation (Do Before the Meeting)

- ☐ Print a one-page "Submission Summary" showing: (a) the 5 evidence folders, (b) the nomination text draft, (c) the YouTube video link, (d) the Google Form URL
- ☐ Test all links one final time in an incognito browser
- ☐ Prepare a 1-page "Teacher Cheat Sheet" with the Google Form link and a bullet-point list of what to paste into each form field
- ☐ Bring a printed copy of the nomination text for the teacher to mark up

15.2 Meeting Agenda

1. Open the evidence folder on a laptop and walk through each subfolder (3 minutes)
2. Show the Impact Report (2 minutes)
3. Play the YouTube video (2.5 minutes)
4. Present the nomination text draft for approval (5 minutes)
5. Confirm: "Are you comfortable submitting this? Do you want to change anything?" (3 minutes)
6. Agree on the submission date: aim for 26–27 March (1 minute)

15.3 If Teacher Requests Changes

Respond within 24 hours. Any changes to the nomination text should be turned around same-day. Any changes to evidence should be implemented and the updated folder link re-shared immediately.

STEP 16: Final QA Checks (24–27 March)

Owner: You | This is a systematic verification pass

Open an **incognito/private browser** for all link testing. Do not rely on cached versions.

16.1 Link Verification

- ☐ Google Drive evidence folder → opens without sign-in
- ☐ Every PDF in the evidence folder → opens and renders correctly
- ☐ Every image file → displays correctly
- ☐ Metrics Log spreadsheet → opens in Google Sheets
- ☐ YouTube video → plays without sign-in (UNLISTED, not Private)
- ☐ Instagram posts → all public and accessible
- ☐ Google Form nomination link → loads correctly: https://forms.gle/kFXV5YbKo63eUtbh9

16.2 Content Verification

- ☐ Impact Report — zero spelling/grammar errors (proofread twice, or have Alex/Grisha proofread)
- ☐ Nomination text — zero errors, teacher has approved
- ☐ All numbers in the Impact Report match the Metrics Log and Mentoring Tracker
- ☐ All student names are anonymised in the Before/After Samples
- ☐ RSC keywords appear in both the Impact Report and nomination text

16.3 Device Testing

- ☐ Open the evidence folder on your phone — do PDFs render?
- ☐ Play the YouTube video on your phone — does it play smoothly?
- ☐ Open the Impact Report on a tablet or different screen size

STEP 17: Submission Day (26–28 March)

Owner: Your teacher (you are the project manager)

SUBMISSION DAY PROTOCOL

1. Be physically present when your teacher submits the Google Form.
2. Have the Teacher Cheat Sheet ready with all links and text pre-copied.
3. Watch them fill in the form. If they need to paste text, have it open on your phone.
4. When they click Submit: immediately screenshot the confirmation page.
5. Send the teacher a thank-you message within 1 hour.
6. Save the confirmation screenshot to the evidence folder (for your own records).
7. Breathe. You have done everything you can.

If anything goes wrong on submission day:

Technical issue with the form → Try a different browser. Try a different device. You have until 31 March.

Teacher is unavailable → This is why you aimed for 26–28 March. You have 3 buffer days.

A link is broken → Fix it immediately. Re-share the updated link. Re-test in incognito.

DAY-BY-DAY EXECUTION CALENDAR

Tear this page out. Put it on your wall. Cross off each day as you complete it.

Date	Tasks	Steps
Wed 4 Mar	Create RSC_Evidence folder structure. Start Metrics Log. Begin Step 8 (Mentoring Tracker). Send testimonial collection form. Begin documenting Lead Writer system (Step 9).	6, 8, 9
Thu 5 Mar	Continue populating Metrics Log with all post data. Write infographic content brief for Nana (Step 10). Screenshot any remaining Instagram posts.	6, 10
Fri 6 Mar	Complete Mentoring Impact Tracker. Start creating Before/After comparison PDFs. Nana begins infographic design. Emergency workshop if not yet delivered.	7, 8, 10
Sat 7 Mar	Finish Before/After samples (3 students). Compile student testimonials. Complete Lead Writer system documentation + process diagram.	8, 9
Sun 8 Mar	Review infographic first draft from Nana. Verify all equations and facts. Final account-level Instagram screenshots.	6, 10
Mon 9 Mar	Finalise all Step 6–9 outputs. Infographic accuracy review. Complete any outstanding evidence capture.	6–9
Tue 10 Mar	Infographic finalised. Print A3 copies. Deliver to Chemistry teacher. Post on Instagram. BEGIN Impact Report drafting (Step 11).	10, 11
Wed 11 Mar	Impact Report: Write Pages 1–2 (Executive Summary, Category Mapping, Evidence of Impact).	11
Thu 12 Mar	Impact Report: Write Page 3 (Institutional Legacy). Screenshot infographic Instagram analytics. First complete draft done.	11
Fri 13 Mar	Proofread Impact Report. RSC keyword check. Export as PDF. Begin evidence folder compilation (Step 12).	11, 12
Sat 14 Mar	Compile evidence folder: Digital Outreach + Workshop folders. Begin drafting nomination text (Step 13).	12, 13
Sun 15 Mar	Compile evidence folder: Mentoring + Teaching Resource + Foundational Documents. Continue nomination text. Begin planning video shoot (Step 14).	12, 13, 14
Mon 16 Mar	Finalise evidence folder. All files checked. Sharing permissions set. Test all links incognito.	12
Tue 17 Mar	Finalise nomination text draft (400–500 words). Film B-roll footage for video.	13, 14
Wed 18 Mar	Send nomination text to teacher for review. Film main video segments (direct to camera + B-roll).	13, 14
Thu 19 Mar	Edit video. Add text overlays, branding, cuts between segments.	14

Fri 20 Mar	Finalise video. Upload to YouTube (Unlisted). Test link. Save link to evidence folder.	14
Sat 21 Mar	Buffer day. Fix anything. Re-test everything.	—
Mon 22 Mar	Schedule teacher review meeting for today or tomorrow.	15
Tue 23 Mar	Teacher review meeting. Walk through evidence folder + nomination text + video.	15
Wed 24 Mar	Implement any teacher-requested changes. Begin final QA.	15, 16
Thu 25 Mar	Full QA pass: all links, PDFs, video, permissions, content verification.	16
Fri 26 Mar	TARGET SUBMISSION DAY. Teacher submits Google Form. Screenshot confirmation.	17
Sat 27 Mar	Backup submission day if Friday was not possible.	17
Sun 28 Mar	HARD DEADLINE for submission. If not submitted yet, this is your last day.	17

Final Note: What Separates Gold from Silver

Silver entries will have one or two categories covered decently. They will show some evidence of chemistry engagement. The teacher will write a generic recommendation.

Gold entries will have four categories covered comprehensively with quantified evidence. They will tell a coherent narrative. The nomination text will be specific, data-driven, and keyword-optimised. The evidence folder will be organised like a legal exhibit. And the video will make judges feel something.

You have built the infrastructure. The 48 foundational documents, the 27-member team, the editorial pipeline, the Instagram presence, the workshop, the mentoring system — all of it exists. Your remaining job is **packaging and presentation**. Make the judges see what you have built in 10 minutes of reviewing your submission.

YOUR NEXT IMMEDIATE ACTION

Today (4 March): Create the RSC_Evidence folder structure in Google Drive. Start the Metrics Log. Send the testimonial collection form to students.

Tonight: Write the infographic content brief and send to Nana.

Tomorrow (5 March): Begin the Mentoring Impact Tracker spreadsheet.

Do not wait. 27 days. The clock is running.

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